

SONiC SmartFabric Manager for PowerFlex Deployment Guide

Abstract

This guide covers using Dell Technologies SONiC SmartFabric Manager to configure a PowerSwitch leaf-spine topology for single rack or multirack PowerFlex deployments.

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1 Introduction

This deployment guide covers Dell Technologies PowerSwitch configuration using SmartFabric Manager for SONiC (SFM) for single rack or multirack greenfield PowerFlex deployments.

SFM automates the configuration of the leaf-spine fabric that is used in this deployment. The leaf-spine fabric uses EVPN VXLAN, which enables a virtual Layer 2 overlay network to run over a physical Layer 3 underlay network. Use of a VXLAN leaf-spine fabric is a best practice in the data center.

1.1 Enterprise SONiC Distribution by Dell Technologies

SONiC is an open-source, Linux-based network operating system that runs on switches from multiple vendors and ASICs. Enterprise SONiC Distribution by Dell Technologies is a hardened and validated version of SONiC that is supported on certain Dell PowerSwitch platforms. Dell Enterprise SONiC supports an intuitive command-line interface, object-based administration through a REST interface, Google's gRPC Network Management Interface (gNMI), and SmartFabric Manager.

1.2 SmartFabric Manager for SONiC

SmartFabric Manager for SONiC (SFM) simplifies network setup and maintenance by automating network deployment and providing an easy-to-use web user interface.

SFM is a virtual machine that runs on VMware ESXi, provided in OVA format, or on Linux KVM, provided in QCOW2 format. SFM configures Dell PowerSwitch systems using REST API commands and offers customizable blueprints to simplify configuration for different solutions, including PowerFlex.

The PowerFlex deployment in this guide uses a Border Gateway Protocol (BGP) Ethernet VPN (EVPN) with Virtual Extensible LAN (VXLAN) leaf-spine fabric. Use of a VXLAN leaf-spine fabric is a best practice in the data center. Manually configuring the switches for this fabric would require hundreds of CLI commands. SFM automates the fabric creation, making it fast, easy, and with a much lower probability of errors than configuring the fabric manually.

SFM comes with a 180-day free trial for up to 10 switches. If your PowerFlex deployment uses more than 10 switches in leaf or spine roles, you must buy a license before proceeding. See the *SFM User Guide* for more information about licensing.

1.3 PowerFlex

PowerFlex is a highly scalable compute and storage infrastructure that integrates PowerFlex software-defined storage software with Dell PowerEdge servers. PowerFlex supports a small cluster starting point of four nodes, bring-your-own networking hardware, and bring-your-own management hardware.

PowerFlex allows growth and management of compute and storage resources together or independently depending on business needs. PowerFlex uses PowerFlex Manager for management and operations. PowerFlex Manager enables users to build, automate, and simplify implementation, expansion, and life cycle management.

This deployment uses PowerFlex partial network automation (PNA). With PNA, PowerFlex manager automatically configures the VLANs and IP addresses on the PowerFlex nodes, but it does not configure the switches.

Note: PowerFlex full network automation (FNA) is not supported with Dell Technologies Enterprise SONiC or SFM.

1.4 Typographical conventions

The CLI and UI examples in this document use the following conventions:

Text Style	Representation
Monospace Text	CLI examples
Bold Monospace Text	• CLI examples which wrap the page • To highlight information in CLI output
<i>Italic Monospace Text</i>	Variables in CLI examples
Bold text	• UI fields • Information which is entered in the UI

1.5 We value your feedback

Dell Technologies and the authors of this document welcome your feedback on the solution and the solution documentation. Contact the Dell Technologies Solutions team by [email](#).

Authors: Dell Technologies Network Solutions Engineering, Technical Marketing, and Information Design and Development teams

2 Hardware used in this guide

This section briefly describes the Dell PowerSwitch and PowerFlex hardware that is used for this PowerFlex on SONiC deployment. [Validated components](#) contains a complete listing of all hardware and software used. [Supported Dell PowerSwitch systems](#) provides a complete list of switches that are supported for this deployment.

2.1 Dell PowerSwitch S5248F-ON

The Dell PowerSwitch S5248F-ON is a 1-Rack Unit (1U), multilayer switch with 48x 25 GbE SFP28, 4x 100 GbE QSFP28, and 2x 200 GbE QFSP28-DD ports. This guide uses two S5248F-ON switches in each rack as leaf switches.



Figure 1 Dell PowerSwitch S5248F-ON

2.2 Dell PowerSwitch S5232F-ON

The Dell PowerSwitch S5232F-ON is a 1U, multilayer switch with 32x 100 GbE QSFP28 ports and 2x 10 GbE SFP+ ports. This guide uses four S5232F-ON switches as spine switches.



Figure 2 Dell PowerSwitch S5232F-ON

2.3 Dell PowerSwitch S3248T-ON

The Dell PowerSwitch S3248T-ON is a 1U switch with 48x 1 GbE BASE-T ports, 4x 10 GbE SFP+ ports, and 2x 100 GbE QSFP28 ports. This guide uses one S3248T-ON switch in each rack for out-of-band (OOB) management traffic.



Figure 3 Dell PowerSwitch S3248T-ON

Note: Other Dell switches, such as the N3248TE-ON, or third-party switches with 1 GbE BASE-T ports and 10 GbE or faster uplink ports can also be used for OOB management traffic.

2.4 Dell PowerFlex appliances and custom nodes

PowerFlex R640, R650, or R660 appliances and custom nodes are supported for this deployment. PowerFlex appliances and custom nodes are built on Dell PowerEdge servers. Each PowerFlex node in this deployment requires two 25 GbE dual-port SFP28 adapters and an iDRAC. PowerFlex nodes used as PowerFlex management controllers must also have a 10 GbE or faster NIC available to connect to the OOB management switch. PowerFlex compute nodes with three 25 GbE dual-port SFP28 adapters are also supported.



Figure 4 Dell PowerFlex node

3 Topology

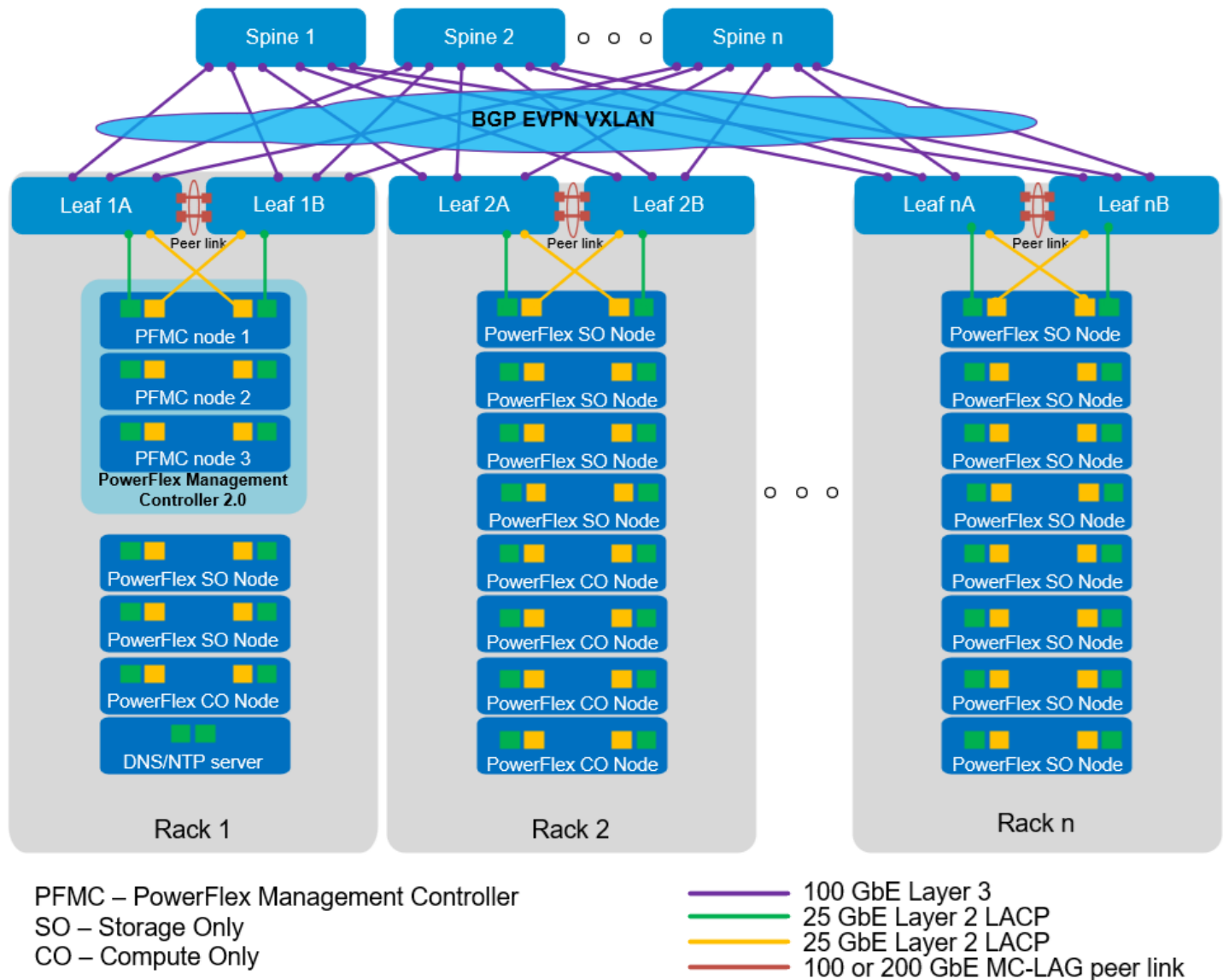
The topology for this deployment is divided into two parts:

1. Production, or in-band
2. Out-of-band (OOB) management

The production topology contains redundant components and is used for all mission-critical and end-user network traffic. The OOB management network is an isolated network for hardware management.

3.1 Production topology

The production topology that is used in this guide is shown in the figure below.



Each rack has two leaf switches connected to each other with a multichassis link aggregation group (MC-LAG) peer link. Downlinks from leaf switches to devices in the rack use LACP with MC-LAG. Every leaf switch is connected to every spine switch to form a leaf-spine topology.

Note: Leaf switches are sometimes called top-of-rack (ToR) switches.

In this deployment example, a three-node PowerFlex Management Controller (PFMC) 2.0 cluster is in Rack 1. The remaining nodes in Racks 1-n are compute only (CO) and storage only (SO) nodes. This deployment uses four 25 GbE network adapter ports per PowerFlex node for production network connections, as shown in the figure above.

Note: The topology used to validate this guide consisted of four racks, eight leaf switches, and four spine switches.

BGP EVPN with VXLAN is used to stretch Layer 2 networks across the Layer 3 leaf-spine fabric. Equal-Cost Multi-Path routing (ECMP) is enabled on the switches to use all available bandwidth on the leaf-spine connections. SFM automates the configuration of this fabric.

The DNS/NTP server that is used for PowerFlex name resolution and time synchronization can be located anywhere that is reachable from the PowerFlex Node Management network. In this example, it is in Rack 1 and has one connection to each leaf in the rack for redundancy.

3.2 Production topology connection details

Production network connection details for this deployment example are shown in the figure below.

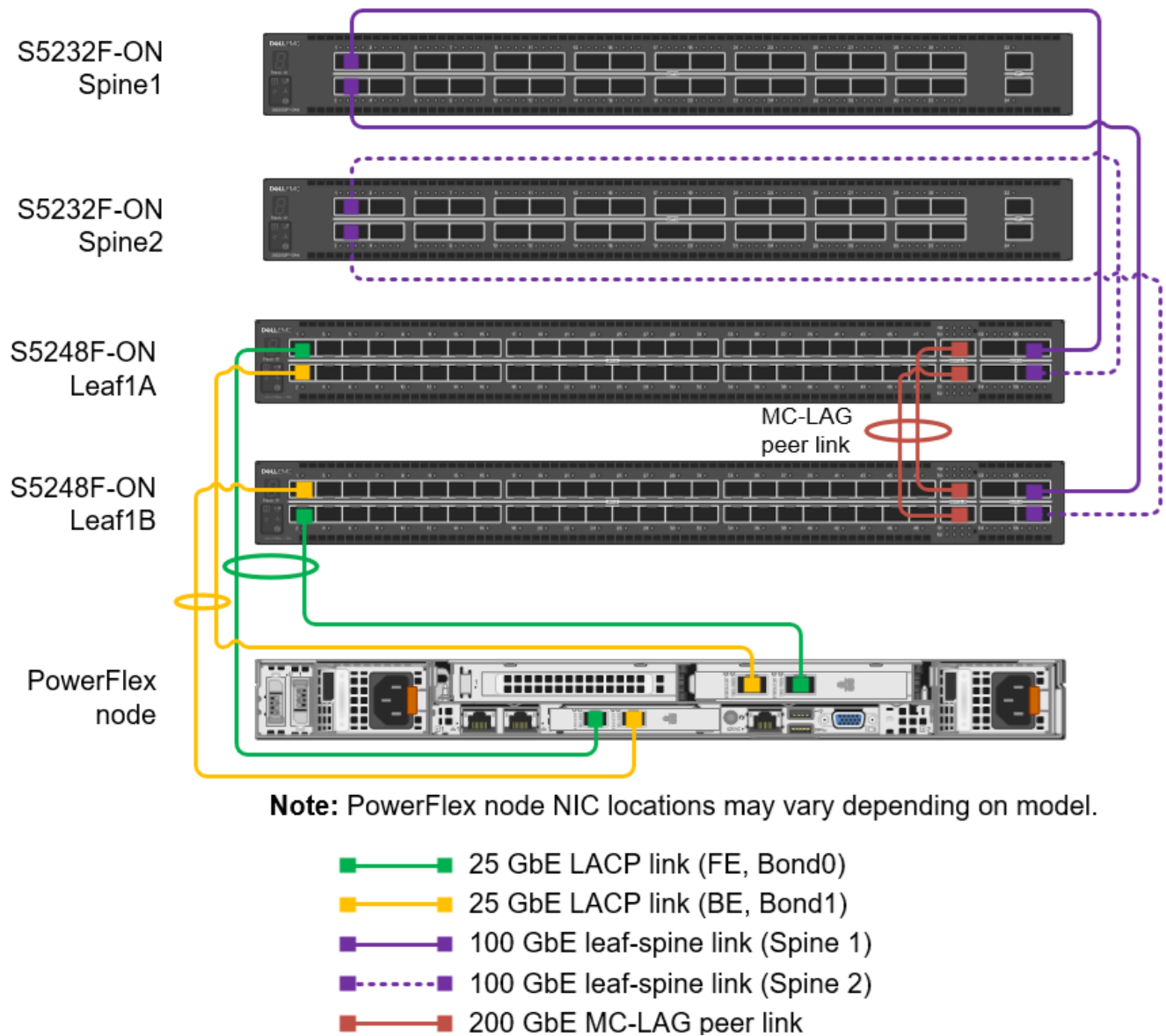


Figure 6 Production network connection details

Note: Other PowerFlex nodes, leaf switches, and spine switches that are not shown are connected in the same manner. PowerFlex node NIC locations may vary depending on the model.

Each leaf switch has one connection to each spine switch. For this deployment, all leaf-spine connections are 100 GbE.

All PowerFlex nodes have two connections to each leaf. The connections in this example are 25 GbE. The four 25 GbE connections from each PowerFlex node to the leaf switches are configured in two LACP link aggregation groups (LAGs). The LAGs are named Front End (FE) and Back End (BE) on the PFMC nodes and Bond0 and Bond1 on the CO and SO nodes.

For the MC-LAG peer link connections, two QSFP28-DD double density ports (2x 100 GbE interfaces per physical port), are available on S5248F-ON switches. They are used to create a 400 GbE port channel that is the MC-LAG peer link. This connection requires QSFP28-DD DAC cables or optics.

Note: On leaf switches without QSFP28-DD ports, QSFP28 100 GbE or SFP28 25 GbE ports are used for peer link connections. Dell Technologies recommends using at least two physical ports on each leaf switch for the peer link. If 25 GbE ports are used, Dell Technologies recommends using at least four connections on each leaf for sufficient bandwidth. Multiple peer link connections provide redundancy and additional bandwidth if a switch is taken offline or if there is a failure. The peer link is used for data traffic only if there is a link failure that requires the peer link to reach the destination.

3.3 OOB management topology

The out-of-band (OOB) management network is an isolated network for remote management of hardware. This hardware includes PowerFlex nodes and PowerSwitch systems. It can also include other devices using their dedicated management ports.

One S3248T-ON switch is typically installed in each rack for OOB management network connections, as shown in the following figure. Other Dell or third-party switches with 1 GbE BASE-T ports can also be used for OOB management traffic.

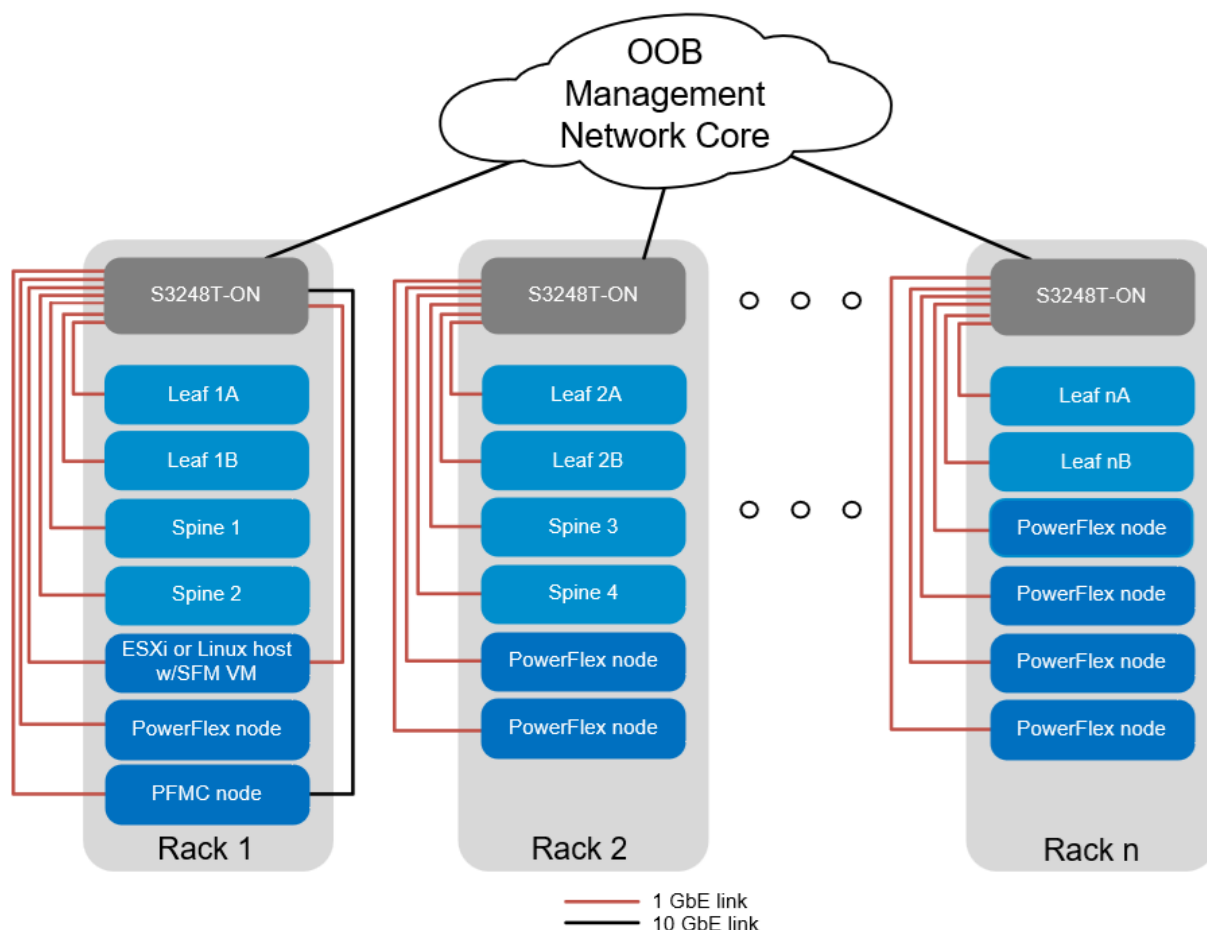


Figure 7 OOB management network topology

Four 10 GbE SFP+ ports are available on each S3248T-ON. These ports are typically used as downlinks to a 10 GbE NIC on each PFMC node and as uplinks to the OOB management network core. There are also two 100 GbE QSFP28 ports available on each S3248T-ON if needed.

The 1 GbE 1000BASE-T ports on each S3248T-ON are connected downstream to hardware management ports on each device in the rack. Those ports include the PowerSwitch management ports and PowerFlex node iDRAC ports. Dedicated management ports on other devices in the rack can also be connected to the OOB management switches.

SFM uses the OOB management network to configure and manage the leaf and spine switches that form the production topology. The ESXi or Linux host with the SFM VM must be accessible on this network. For this reason, the host containing SFM requires a 1 GbE NIC connected to the OOB management network in addition to its iDRAC port.

OOB management switch configuration is not detailed in this guide. The OOB management network can be Layer 2 (switched) or Layer 3 (routed) and configured as needed to suit your environment.

Note: For reference, devices on the OOB Management network in this guide use the 100.67.0.0/16 IP address range. All addresses that are used in this guide are examples only.

3.4 OOB management connection details

All PowerSwitch management ports and PowerFlex node iDRACs are connected to the OOB management switch in each rack as shown below.

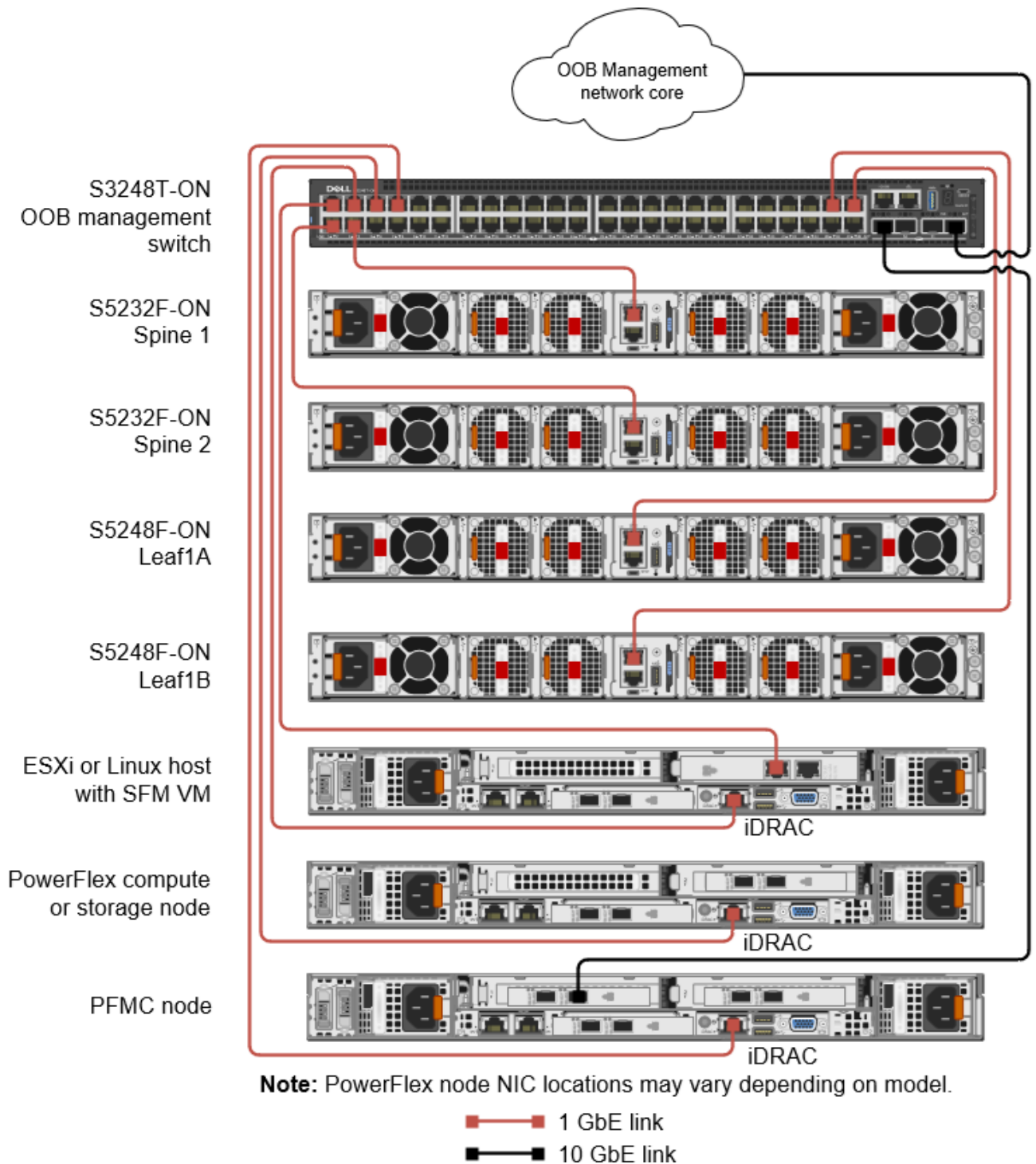


Figure 8 OOB management network connection details

The ESXi or Linux host with the SFM VM has two connections to the OOB management network: the 1 GbE iDRAC and an additional 1 GbE NIC for SFM to communicate with the PowerSwitch leaf and spine systems.

The PowerFlex compute and storage nodes only require the iDRAC connection to the OOB network. Each PFMC node has two connections to the OOB management network: the 1 GbE iDRAC and an additional 10

GbE NIC. PowerFlex Manager uses the PFMC node's 10 GbE interface to communicate with the iDRACs on the PowerFlex compute and storage nodes.

Note: Other PowerFlex nodes, spines, and leaf switches that are not shown are connected in the same manner. NIC and iDRAC locations may vary depending on the model.

4 Deployment Planning

This chapter covers items to consider before deploying a PowerFlex network with SFM.

4.1 Minimum networking requirements

Note: In this guide, “single rack” means topologies with only two leaf switches. “Multirack” means topologies with four or more leaf switches.

The minimum networking requirements for a PowerFlex on Dell Technologies Enterprise SONiC deployment with SFM include the following:

- SFM version 1.2 or later. SFM comes with a 180-day free trial for up to 10 switches. If your PowerFlex deployment uses more than 10 switches in leaf or spine roles, you must purchase a license before proceeding. See the *SFM User Guide* for more information about licensing.
- Single rack deployments require two supported Dell PowerSwitch leaf switches of the same model.
- Multirack deployments require two supported Dell PowerSwitch leaf switches of the same model per rack, plus at least two supported Dell PowerSwitch spine switches.

Note: For a new deployment that is done using the SFM PowerFlex blueprint as shown in this guide, all leaf switches in the initial fabric must be the same model. All spine switches in the initial fabric must also be the same model. The fabric can later be expanded using other supported PowerSwitch models.

- All Dell PowerSwitch leaf and spine systems in the fabric must be running the same version of Dell Technologies Enterprise SONiC.
- Dell Technologies Enterprise SONiC 4.4.1 or later is required. An Enterprise Standard or Enterprise Premium bundle is required for leaf and spine switches in this deployment.
- One 1 GbE BASE-T switch for OOB management connections is required. Dell Technologies recommends using one PowerSwitch S3248T-ON per rack.
- One DNS server and one NTP server that are reachable from the PowerFlex Node Management network are required.
- Each PowerFlex node requires four 25 GbE SFP28 connections to the leaf switches.
- Each leaf to spine connection requires at least one 100 GbE QSFP28 connection. Provide enough connections to ensure that the oversubscription ratio at the leaf switches does not exceed 2:1.
- MC-LAG peer links between leaf switch pairs require at least two connections for fault tolerance. Each link can be 25 GbE SFP28, 100 GbE QSFP28, or 200 GbE QSFP28-DD (available on S5248F-ON). If 25 GbE ports are used, Dell Technologies recommends using at least four connections for sufficient bandwidth.
- One 1 GbE (1000BASE-T) connection is required for each switch OOB management port.
- One 1 GbE (1000BASE-T) connection is required for each PowerFlex node iDRAC port.
- One 10 GbE or faster connection is required for each PFMC node to access the OOB management network in addition to the PFMC node's 1 GbE iDRAC port.
- On server running ESXi 7.0 or later or Linux KVM to host the SFM VM. The server must have at least one network connection that can reach the OOB management network. This connection is used for SFM to communicate with the leaf and spine switches. The SFM VM requires 32 GB RAM, 120 GB free disk space, and 16 virtual CPUs.

Note: See your PowerFlex documentation for additional PowerFlex node requirements.

4.2 Maximum deployment size

The maximum number of PowerSwitch systems and PowerFlex nodes that are supported for a PowerFlex on SONiC leaf-spine fabric deployment, with or without SFM, are:

- Spine switches - 6
- Leaf switches - 32
- PowerFlex nodes - 352 (22 nodes per leaf pair)

4.3 Supported Dell PowerSwitch systems

Dell PowerSwitch systems that are supported as leaf or spine switches for a PowerFlex on Dell Technologies Enterprise SONiC deployment, with or without SFM, are shown in the table below. If you are using an SFM blueprint to configure the switches, all leaf switches in the fabric must be the same model during the initial deployment. The fabric can be expanded after initial deployment to include other supported leaf switch models.

Note: The number of 100 GbE QSFP28 ports available on the leaf switches limits the maximum number of spine switches in this deployment.

Table 1 Supported switches and roles

Switch	Ports	Role
S5212F-ON	12x 25 GbE SFP28 3x 100 GbE QSFP28	Leaf
S5224F-ON	24x 25 GbE SFP28 4x 100 GbE QSFP28	Leaf
S5248F-ON	48x 25 GbE SFP28 2x 200 GbE QSFP28-DD (See note below.) 4x 100 GbE QSFP28	Leaf
S5232F-ON	32x 100 GbE QSFP28 2x 10 GbE SFP+	Spine

Note: If needed, QSFP28-DD ports on S5248F-ON switches can be used in 100 GbE mode with QSFP28 connectors for additional leaf-spine connections.

For OOB management, Dell Technologies recommends one PowerSwitch S3248T-ON per rack. Other Dell or third-party switches with 1 GbE BASE-T ports and 10 GbE or faster uplink ports can also be used for OOB management traffic.

4.4 VLANs and IP addresses

In this deployment, traffic is divided into different networks as shown in the *Dell PowerFlex Appliance with PowerFlex 4.x Network Planning Guide*. The VLANs required may vary based on your deployment. Depending on the traffic type, some networks are Layer 2 (no gateway), and some are Layer 3 (with a gateway). The management and site replication networks must be accessible from devices on other networks, so they are Layer 3. The remaining networks are Layer 2.

Note: The VLAN IDs and IP addresses used in this guide are examples only and can be changed as needed. VLAN IDs 1-3446, 3939, and 3961-3966 are available for use. SFM reserves the remaining VLAN IDs.

Table 2 VLAN IDs and network addresses

VLAN ID	Description	Network Name	Network	Gateway
103	vCenter high availability	flex-vcsa-ha-L2	192.168.103.0/24	None
105	Node Management	flex-node-mgmt-L3	192.168.105.0/24	192.168.105.254
106	Compute vMotion	flex-vmotion-L2	192.168.106.0/24	None
140	PFMC Management	pfmc-sds-mgmt-L3	192.168.140.0/24	192.168.140.254
141	PFMC Data 1	pfmc-sds-data1-L2	192.168.141.0/24	None
142	PFMC Data 2	pfmc-sds-data2-L2	192.168.142.0/24	None
143	PFMC vMotion	pfmc-vmotion-L2	192.168.143.0/24	None
150	Storage Management	flex-stor-mgmt-L3	192.168.150.0/24	192.168.150.254
151	Storage Data 1	flex-data1-L2	192.168.151.0/24	None
152	Storage Data 2	flex-data2-L2	192.168.152.0/24	None
161	Site Replication Traffic 1	flex-repl1-L3	192.168.161.0/24	192.168.161.254
162	Site Replication Traffic 2	flex-repl2-L3	192.168.162.0/24	192.168.162.254

4.5 LAG to VLAN mapping

Each PowerFlex node is a member of several different VLANs depending on its type. VLAN membership by node type is shown in the table below.

Table 3 VLAN membership by node type

PowerFlex Node Type	VLAN membership
PowerFlex Management Controller (PFMC)	103, 105, 140-143, 150
Compute Only (CO)	105-106, 151-152
Storage Only (SO)	150-152, 161-162

The four network connections from each PowerFlex node to the two leaf switches are configured in two LACP LAGs as shown in [Production network connection details](#). On PFMC nodes running ESXi, the LAGs are named Front End (FE) and Back End (BE). On PowerFlex CO or SO nodes running the intelligent catalog embedded Linux operating system, the LAGs are named Bond0 and Bond1.

The VLANs are distributed between the two LAGs as shown in the table below.

Table 4 VLAN LAG assignments by node type

PowerFlex Node Type	LAG 1: FE or Bond0	LAG 2: BE or Bond1
PFMC	105, 140, 150	103, 141-143
CO	105-106	151-152
SO	150-151, 161	152,162

5 Use SFM to create and verify the fabric

This section shows how to use SFM to create and verify the fabric for this PowerFlex deployment example. For other SFM features and additional information, see the *SFM User Guide*.

5.1 Prepare switches for SFM

Dell PowerSwitch systems that are used in leaf or spine roles in this deployment must be running Dell Technologies Enterprise SONiC 4.4.1 or later. An Enterprise Standard or Enterprise Premium bundle is required.

5.1.1 Set a password and configure the management interface

For each Dell PowerSwitch that will be a leaf or a spine in the fabric, do the following:

1. Connect to the switch using the console serial port.
2. Log in to the CLI as `admin`.
3. Change the admin password from the default, `YourPaSsWoRd`, to a secure password during the first login when prompted.
4. Configure the OOB management interface, named Management 0.
SFM uses the management interface to communicate with the leaf and spine switches. For SFM, Dell Technologies recommends using a static IP address instead of DHCP on Management 0 to ensure that it does not change. Use a unique IP address on each switch. If routing is used on your OOB management network, also specify the gateway address as shown in the configuration example below.

```
admin@sonic:~$ sonic-cli
sonic# configure
sonic(config)# interface Management 0
sonic(config-if-Management0)# ip address 100.67.100.189/24 gwaddr 100.67.100.254
sonic(config-if-Management0)# end
sonic# write memory
```

Note: If any leaf or spine switches have previously been configured (other than changing the admin password and configuring Management 0), run the `write erase` command and reboot each applicable switch. The admin password and Management 0 interface configuration are retained. All other configuration settings on the switch are reset to the factory defaults. The switches must be in this state for the SFM blueprint to be applied successfully.

5.1.2 Connect the switches and SFM host

Make the switch and SFM host connections as shown in [Production network connection details](#) and [OOB management network connection details](#). The PowerFlex nodes do not have to be connected now.

5.2 Deploy SFM

Deploy the SFM virtual machine to a VMware ESXi or Linux host as shown in the *SFM User Guide*. SFM is available in OVA format for ESXi or in QCOW2 format for QEMU/KVM on Linux. The host must have network connectivity to the OOB management network so that SFM can communicate with the leaf and spine switches. Example network connections are shown in [OOB management network connection details](#).

Note: SFM comes with a 180-day free trial for up to 10 switches. If your PowerFlex deployment uses more than 10 switches in leaf or spine roles, you will need to purchase a license before proceeding. See the *SFM User Guide* for more information on licensing.

5.3 Export the PowerFlex blueprint

Export the PowerFlex blueprint from SFM as follows:

1. In a browser, connect to the SFM UI and log in.
2. In the left pane, select **Blueprints**. On the **Blueprints** page, a table of blueprints displays.
3. Click the **Blueprint Type** heading to sort by type.
The **Converged Infrastructure** blueprint type includes two PowerFlex blueprints:
 - 4x2 PowerFlex leaf S5248F-ON and spine S5232F-ON (for a four leaf/two spine deployment)
 - 8x4 PowerFlex leaf S5248F-ON and spine S5232F-ON (for an eight leaf/four spine deployment).
4. Select the PowerFlex blueprint that most closely matches the number of leaf and spine switches that are used in your deployment.
The number of switches and models that are used in the blueprint name do not need to match your topology exactly. You will edit the blueprint to match your topology in subsequent steps.

	Name	Blueprint Type	Type	No. of Switches
☐	8x4 Leaf Z9432F-ON spine Z9664F-ON	Access	PreCanned	12
☐	Pod based 3Tier	Access	PreCanned	16
☐	2x2 Leaf Z9432F-ON spine Z9664F-ON	Access	PreCanned	4
☐	2x2 Leaf Z9432F-ON spine Z9664F-ON	Converged Infrastructure	PreCanned	4
☐	4x2 PowerFlex leaf S5248F-ON and spine S5232F-ON	Converged Infrastructure	PreCanned	6
☐	8x4 Leaf Z9432F-ON spine Z9664F-ON	Converged Infrastructure	PreCanned	12
☒	8x4 PowerFlex leaf S5248F-ON and spine S5232F-ON	Converged Infrastructure	PreCanned	12

Figure 9 PowerFlex blueprint selected

5. Click **EXPORT**.
6. In the confirmation dialog box, click **EXPORT** to download the blueprint.
The blueprint is downloaded to your workstation in ZIP format.
7. Extract the blueprint ZIP file.
The ZIP file contains two files in XLSX format:
 - *Fabric_size_PowerFlex_leaf_S5248F_ON_and_spine_S5232F_ON.xlsx*
 - *Fabric_size_PowerFlex_leaf_S5248F_ON_and_spine_S5232F_ON_spec.xlsx*

The first file is known as the “blueprint file.” It is for fabric deployment and is covered in detail in this chapter. The second file is known as the “spec file.”

Note: The spec file is not used in this deployment. See the *SFM User Guide* for more information about the spec file.

5.4 Edit the Blueprint

This section shows how to edit the blueprint for your PowerFlex deployment. When editing the blueprint, you can add or delete rows, except the heading row, as needed. Never add or delete columns or modify the headings. Open the blueprint XLSX file (not the spec file) in a spreadsheet application such as Microsoft Excel.

Note: Some heading cells have a red triangle in the upper right corner of the cell. You can hover your mouse pointer over the triangle for more information about cell formatting.

The blueprint file has 12 tabs. The following sections begin with a screenshot of each completed tab for this deployment. Below the screenshot is a list of requirements for each column. Data is required in each column except where noted as optional.

5.4.1 Overview tab

The first tab in the blueprint file is named **Overview** and is shown in the figure below.

	A	B	C	D	E	F	G	H	I	J
1	Name	Description	late V	Namespace	Topology Type	Leaf Node Model	Spine Node Model	Network Ty	Border Leaf Node Model	Super Spine Node Model
2	Example 8x4 PowerFlex leaf S5248F-ON	Access Layer 2 Fabric	1.0.0	Converged It RackBasedClos	S5248F-ON	S5232F-ON	Bridged			

Figure 10 Overview tab

- A. **Name** – This cell contains the name that is displayed in the SFM blueprint list. Edit the name to be unique in SFM. If it is not unique, SFM blocks the import to prevent overwriting the original template. For this example, the name is modified by adding “**Example**” is added to the beginning of the name.
- B. **Description** (optional) – Edit this cell if needed.
- C. **Template Version** – This cell is protected and must not be changed.
- D. **Namespace** – This cell is protected and must not be changed.
- E. **Topology Type** – This cell is protected and must not be changed.
- F. **Leaf Node Model** – If you are not using the default leaf switch model, S5248F-ON, enter the leaf switch model that you are using. All leaf switches in the initial deployment must be the same model. Supported leaf switches are shown in [Supported switches and roles](#).
- G. **Spine Node Model** – The only switch supported as a spine for PowerFlex on SONiC deployments is S5232F-ON. For a two-leaf fabric without spines, this cell is ignored.
- H. **Network Type** – This cell is protected and must not be changed.
- I. **Border Leaf Node Model** – Leave this cell blank. This cell is for future use.

Note: Any leaf pair in this PowerFlex topology may be used as border leaves by creating uplinks to external switches in the SFM UI after the fabric is created.

- J. **Super Spine Node Model** – Leave this cell blank. Super spines are not supported in this PowerFlex deployment.

5.4.2 Nodes tab

The **Nodes** tab is prepopulated with the number of leafs and spines that are used in the blueprint name. Add or delete rows as needed to match the number of leafs and spines that are used in your fabric.

	A	B	C	D	E	F	G	H	I	J	K	L
1	Node Reference Key	Node Role	Node Name	IP Address	Username	Password	Pod Name	Rack Name	Service Tag	MAC Address	Spine Id	Super Spine Id
2	leaf1a	Leaf	leaf1a	100.67.124.189	admin	P@ssw0rd	1	1				
3	leaf1b	Leaf	leaf1b	100.67.124.190	admin	P@ssw0rd	1	1				
4	leaf2a	Leaf	leaf2a	100.67.124.187	admin	P@ssw0rd	1	2				
5	leaf2b	Leaf	leaf2b	100.67.124.188	admin	P@ssw0rd	1	2				
6	leaf3a	Leaf	leaf3a	100.67.124.191	admin	P@ssw0rd	1	3				
7	leaf3b	Leaf	leaf3b	100.67.124.192	admin	P@ssw0rd	1	3				
8	leaf4a	Leaf	leaf4a	100.67.124.193	admin	P@ssw0rd	1	4				
9	leaf4b	Leaf	leaf4b	100.67.124.194	admin	P@ssw0rd	1	4				
10	spine1	Spine	spine1	100.67.124.185	admin	P@ssw0rd	1				1	
11	spine2	Spine	spine2	100.67.124.186	admin	P@ssw0rd	1				2	
12	spine3	Spine	spine3	100.67.124.195	admin	P@ssw0rd	1				3	
13	spine4	Spine	spine4	100.67.124.196	admin	P@ssw0rd	1				4	
14												

Figure 11 Nodes tab

For each leaf and spine switch in the fabric, enter the following:

- A. **Node Reference Key** – This cell contains a unique key for each switch. If you change it from the default, Dell Technologies recommends using the same value that you use for the **Node Name** (column C) for simplicity.
- B. **Node Role** – Enter the role, **Leaf**, or **Spine**.
- C. **Node Name** – Enter a unique hostname for the switch.
- D. **IP Address** – Enter the OOB management IP address that is configured on the switch.
- E. **Username** – Enter a username with the admin role on the switch. The default is **admin**.
- F. **Password** – Enter the password for the admin user that is configured on the switch.
- G. **Pod Name** – Enter a numerical value for the pod. If you change it, use the same number for all switches in this deployment.
- H. **Rack Name** – For each leaf switch, enter a rack number. There must be exactly two leafs for each rack number as shown above. Leave this cell blank for spine switches.
- I. **Service Tag (optional)** – If the switch service tag is entered, SFM validates that it matches the service tag of the switch when SFM validates the blueprint.
- J. **MAC Address (optional)** – If the switch MAC address is entered, SFM validates that it matches the MAC address of the switch when SFM validates the blueprint.
- K. **Spine Id** – For each spine switch, enter a unique number.
- L. **Super Spine Id** – Leave this column blank. Super spines are not supported in this PowerFlex deployment.

5.4.3 Breakouts tab (optional)

The **Breakouts** tab is not required. In this PowerFlex deployment, port breakouts should not be used for the leaf-to-spine connections or for the leaf-to-PowerFlex node connections. If breakouts are not needed, this tab can be left blank as is done in this example.

Note: Only the 100 GbE ports on the switches that are listed in [Supported Dell PowerSwitch systems](#) can be broken out using this tab. If you need to change the speed of a 25 GbE port on the S52xx leaf switches, use the SFM UI after deployment. See the *SFM User Guide* for more information.

	A	B	C
1	Node Reference Key	Interface Name	Breakout Mode
2			

Figure 12 Breakouts tab

- A. **Node Reference Key** – Enter the reference key for the switch. The reference key must be identical to the key used on the **Nodes** tab for the switch.
- B. **Interface Name** – Enter the interface name in the format *Eth1/port_number*.
- C. **Breakout Mode** – Enter the breakout mode in the format *number_of_portsXspeedG*. For example, to break out a 100 GbE port to four 25 GbE interfaces, enter **4X25G**.

Add rows to the sheet if you need to break out additional ports.

5.4.4 Fabrics tab

The **Fabrics** tab contains information that applies to the entire fabric.

Name	Fabric Type	Fabric Sub Type	RoCE Enabled	Cut-through Enabled	ARS Enabled	ISL Connection Policy	STP Edge	LACP Fallback Mode	Anycast MAC	Startup Delay	MAC Holdtime	Neighbor Holdtime	EVI Disable	NTP Server	DNS Server	sFlow Collector	Leaf ASN	Spine ASN	Leaf IP	Spine IP	GP CIDR	VTEP CIDR	Site Id
PF-Fabric	VxLan		FALSE	FALSE	FALSE	Auto	Enable	Fallback	00:00:00:11:11:11	300	1080	1080	False				65011	65012	65014	65013	172.16.1.1	172.30.0.1	1

Figure 13 Fabrics tab

- A. **Name** – Enter a name for your fabric.
- B. **Fabric Type** – Do not edit this cell. The fabric type must be **VxLan** for this deployment.
- C. **Fabric Sub Type (optional)** – The current fabric sub type is ERA (Enterprise Reference Architecture). This fabric sub type refers to an NVIDIA type of deployment. If the deployment is not an NVIDIA Enterprise Reference Architecture, then it should be left untouched.
- D. **RoCE Enabled** – If RoCE is needed to ensure a lossless fabric, RoCE should be set to “**True**”.
- E. **Cut-through Enabled** – Cut-through switching is disabled by default.
- F. **ARS Enabled** – Adaptive routing and switching is disabled by default.
- G. **ISL Connection Policy** – The ISL connection policy indicates the connection methodology between the leaf and spine switches. Currently, there are three types of connections: Auto, Ranged, and Fixed.
- H. **STP Edge** – Leave this set to **Enable** for PowerFlex deployments. Spanning tree is used as a precaution only. When the fabric is properly configured, there will be ports blocked by spanning tree.
- I. **LACP Fallback Mode** – Leave this set to **Fallback** for PowerFlex deployments.
- J. **Anycast MAC** – This setting is autoconfigured for MC-LAG deployment types.
- K. **Startup Delay** – The default delay is set to 300. This delay refers to a VTEP device in a multihoming deployment. The delay parameter is configured to prevent traffic loss from being assigned to a VTEP device that is not ready for traffic forwarding.
- L. **MAC Holdtime** – Configure a mac holdtime to wait before aging out the MAC address of a downstream device learned from a multihomed peer VTEP and it has not been used. The default holdtime is set to 1080 seconds
- M. **Neighbor Holdtime** – This setting refers
- N. **EAD EVI Disable** – This setting refers to EVPN Multihoming if it is being deployed in your environment and it is set to “False” by default, it has to be set to “True” if it is going to be deployed.
- O. **NTP Server (optional)** – Enter the FQDN or IPv4 address of the NTP server that is used by the **DNS Server (optional)** – Enter the IPv4 address of the DNS server used by the switches.
- P. **sFlow Collector (optional)** – SFM is the default sFlow collector. To use a different sFlow collector, enter it using the format: *Management:IPv4_address:RemotePort*. Valid ports are **1-65535**. For example, **Management:100.67.1.51:6343**.

- Q. **Log Server (optional)** – SFM is the default syslog server. To use a different syslog server, enter it using the format: *IPv4_address:RemotePort:MessageType:SeverityLevel*. Valid ports are **1-65535**. Valid message types are **Log** or **Event**. Valid severity levels are **Informational, Notice, Error, Debug, Critical, Warning, Alert, or Emergency**. For example, **100.67.1.52:5140:Log:Informational**.
- R. **Leaf ASN** – This cell contains the BGP autonomous system number (ASN) used by all leafs in the fabric. Dell Technologies recommends leaving this set to **65011** unless it conflicts with your environment. Valid values are **1-4294967295**. This ASN cannot be changed once the fabric is deployed.
- S. **Spine ASN** – This cell contains the ASN used by all spines in the fabric. Dell Technologies recommends leaving this set to **65012** unless it conflicts with your environment. Valid values are **1-4294967295**. This ASN cannot be changed once the fabric is deployed. If you are deploying a two-leaf fabric without spines, SFM reserves this ASN for future spines. This ASN cannot be changed once the fabric is deployed.
- T. **Border Leaf ASN** – There are no leafs designated in SFM as border leafs in this deployment. However, SFM reserves this ASN. Dell Technologies recommends leaving this set to **65014** unless it conflicts with your environment. Valid values are **1-4294967295**. This ASN cannot be changed once the fabric is deployed.
- U. **Super Spine ASN** – There are no super spines in this deployment. However, SFM reserves this ASN. Dell Technologies recommends leaving this set to **65013** unless it conflicts with your environment. Valid values are **1-4294967295**. This ASN cannot be changed once the fabric is deployed.
- V. **BGP CIDR** – This cell contains the IP address block in Classless Inter-Domain Routing (CIDR) format that SFM uses for BGP router IDs. Dell Technologies recommends leaving this set to **172.16.128.0/24**, unless it conflicts with your environment. Each switch in the fabric uses one address from this block for the BGP router ID. If you change the block, make sure there is enough address space to allow for future expansion of the fabric. This address block cannot be changed once the fabric is deployed.
- W. **VTEP CIDR** – This cell contains the IP address block in CIDR format that SFM uses for the leaf switch VTEP addresses. Dell Technologies recommends leaving this set to **172.30.0.0/16**, unless it conflicts with your environment. Each leaf switch pair in the fabric consumes one address from this block for the VTEP IP address. If you change the block, make sure there is enough address space to allow for future expansion of the fabric. This address block cannot be changed once the fabric is deployed.
- X. **Site ID** – Leave this set to **1**. The **Site ID** is reserved for future use.

5.4.5 Fabrics ICLLinks tab

The MC-LAG peer link ports on leaf switch pairs are entered on the **Fabric ICLLinks** tab. Add or delete rows as needed to match your deployment. One row is used for each MC-LAG peer link.

	A	B	C	D
1	Source Node Reference Key	Source Interface Name	Destination Node Reference Key	Destination Interface Name
2	leaf1a	Eth1/49	leaf1b	Eth1/49
3	leaf1a	Eth1/50	leaf1b	Eth1/50
4	leaf1a	Eth1/51	leaf1b	Eth1/51
5	leaf1a	Eth1/52	leaf1b	Eth1/52
6	leaf2a	Eth1/49	leaf2b	Eth1/49
7	leaf2a	Eth1/50	leaf2b	Eth1/50
8	leaf2a	Eth1/51	leaf2b	Eth1/51
9	leaf2a	Eth1/52	leaf2b	Eth1/52
10	leaf3a	Eth1/49	leaf3b	Eth1/49
11	leaf3a	Eth1/50	leaf3b	Eth1/50
12	leaf3a	Eth1/51	leaf3b	Eth1/51
13	leaf3a	Eth1/52	leaf3b	Eth1/52
14	leaf4a	Eth1/49	leaf4b	Eth1/49
15	leaf4a	Eth1/50	leaf4b	Eth1/50
16	leaf4a	Eth1/51	leaf4b	Eth1/51
17	leaf4a	Eth1/52	leaf4b	Eth1/52

Figure 14 Fabrics ICLLinks tab

- A. **Source Node Reference Key** – Enter the reference key for the source leaf switch. The reference key must be identical to the reference key used on the [Nodes tab](#) for the switch. In this example, the switch hostname is used as the reference key.
- B. **Source Interface Name** – Enter the MC-LAG peer link source interface in the format *Eth1/port_number*.
- C. **Destination Node Reference Key** – Enter the reference key for the destination leaf switch. The reference key must be identical to the reference key used on the [Nodes tab](#) for the switch. In this example, the switch hostname is used as the reference key.
- D. **Destination Interface Name** – Enter the MC-LAG peer link destination interface in the format *Eth1/port_number*.

5.4.6 Tenants tab

A nondefault Virtual Routing and Forwarding (VRF) for PowerFlex is defined on the Tenants tab.

VRFs allow multiple instances of independent routing tables to co-exist within the same router simultaneously. SFM refers to VRFs as tenants.

Using a nondefault tenant is a best practice because it enables symmetric integrated bridging and routing (IRB) with VXLAN. In symmetric IRB, both the ingress and egress VTEPs route packets to their destination. The alternative is asymmetric IRB, which does not handle routing as efficiently as symmetric IRB.

	A	B
1	Tenant Id	Description
2	PF_VRF	Powerflex VRF
3		

Figure 15 Tenants tab

- A. **Tenant Id** – Enter a VRF name to use for the PowerFlex networks.
- B. **Description (optional)** – Enter a description of the VRF.

5.4.7 Networks tab

The **Networks** tab is prepopulated with the example networks shown earlier in [VLAN IDs and network addresses](#). Add or delete rows as needed to match the number of networks used in your deployment.

	A	B	C	D	E	F
1	Network Name	Tenant Id	Network Type	Vlan	Address Family	Gateway IP Address
2	103-flex-vcsa-ha-L2	PF_VRF	VxLan	103	INET	
3	105-flex-node-mgmt-L3	PF_VRF	VxLan	105	INET	192.168.105.254/24
4	106-flex-vmotion-L2	PF_VRF	VxLan	106	INET	
5	140-pfmc-sds-mgmt-L3	PF_VRF	VxLan	140	INET	192.168.140.254/24
6	141-pfmc-sds-data1-L2	PF_VRF	VxLan	141	INET	
7	142-pfmc-sds-data2-L2	PF_VRF	VxLan	142	INET	
8	143-pfmc-vmotion-L2	PF_VRF	VxLan	143	INET	
9	150-flex-stor-mgmt-L3	PF_VRF	VxLan	150	INET	192.168.150.254/24
10	151-flex-data1-L2	PF_VRF	VxLan	151	INET	
11	152-flex-data2-L2	PF_VRF	VxLan	152	INET	
12	161-flex-repl1-L3	PF_VRF	VxLan	161	INET	192.168.161.254/24
13	162-flex-repl2-L3	PF_VRF	VxLan	162	INET	192.168.162.254/24
14						

Figure 16 Networks tab

- A. **Network Name** – Enter a name for the network.
- B. **Tenant Id** – This cell must match the **Tenant ID** you defined on the Tenants tab.
- C. **Network Type** – This cell must be **VxLan** for this deployment.
- D. **Vlan** – Enter a VLAN ID for the network. VLAN IDs 1-3446, 3939, and 3961-3966 are available for use. SFM reserves the remaining VLAN IDs.
- E. **Address Family** – This cell must be INET for this deployment.
- F. **Gateway IP Address (Optional)** – Enter the gateway address for the network if applicable.

5.4.8 ClusterProfiles tab

The **ClusterProfiles** tab is used to define clusters. In SFM, PowerFlex nodes can be grouped into clusters based on their network configuration. Nodes in the same cluster use the same VLANs, NIC names, and bond names. This method simplifies configuration as a group of VLANs, NICs, and bonds can be assigned to a cluster instead of each individual node.

Note: The same VLAN, NIC, and bond names can be assigned to more than one cluster. VLANs, NICs, and bonds can be edited on a node-by-node basis in the SFM UI anytime after deployment if needed.

PowerFlex nodes are divided into clusters by node type (CO, SO, and PFMC) in this example because each node type shares the same networking information.

	A	B	C	D	E
1	Cluster Name	Description	Multi-tenancy Support	Type	Tenant Id
2	CO_nodes	CO nodes cluster profile	Enable	Custom	PF_VRF
3	SO_nodes	SO nodes cluster profile	Enable	Custom	PF_VRF
4	PFMC_nodes	PFMC nodes cluster profile	Enable	Custom	PF_VRF
5					

Figure 17 ClusterProfiles tab

- A. **Cluster Name** – Enter a name for the cluster.
- B. **Description (optional)** – Enter a cluster description.
- C. **Multi-tenancy Support** – Leave this set to **Enable**.
- D. **Type** – Leave this set to **Custom**.
- E. **Tenant Id** – Enter the **Tenant Id** that is defined on the [Tenants tab](#) for the PowerFlex networks.

5.4.9 ClusterProfilesNics tab

The **ClusterProfilesNics** tab is used to name NICs. Each PowerFlex node NIC port connected to a leaf switch is assigned a name. The NIC name is used to map the network connections on subsequent tabs. Each PowerFlex node uses four NICs, so there are four rows that are used for each Cluster.

	A	B	C	D
1	Cluster Name	NIC Name	Untagged Network	Tagged Networks
2	PFMC_nodes	NIC0:1 FE		
3	PFMC_nodes	NIC2:1 FE		
4	PFMC_nodes	NIC0:2 BE		
5	PFMC_nodes	NIC2:2 BE		
6	CO_nodes	NIC0:1 B0		
7	CO_nodes	NIC3:1 B0		
8	CO_nodes	NIC0:2 B1		
9	CO_nodes	NIC3:2 B1		
10	SO_nodes	NIC0:1 B0		
11	SO_nodes	NIC3:1 B0		
12	SO_nodes	NIC0:2 B1		
13	SO_nodes	NIC3:2 B1		
14				

Figure 18 ClusterProfilesNics tab

- A. **Cluster Name** – Specify the cluster name from the [ClusterProfiles tab](#). Use one row for each cluster's NIC port that is connected to a leaf switch.
- B. **NIC Name** – Use a unique name for each NIC in a cluster. The same NIC names may be used in other clusters. This example uses the format *NICslot:port <space>bond_name*. You do not have to use this format, but your NIC names must be consistent on subsequent tabs as shown in the following sections.
- C. **Untagged Network** – Leave this column blank. Since this deployment uses LACP bonds, the networks are defined on the [ClusterProfilesBonds tab](#).
- D. **Tagged Networks** – Leave this column blank for the same reason.

5.4.10 ClusterProfilesBonds tab

The **ClusterProfilesBonds** tab is used to create the LACP bonds and assign the applicable networks to each bond.

	A	B	C	D	E	F	G
1	Cluster Name	Type	Mode	NICs	Bond Name	Untagged Network	Tagged Networks
2	PFMC_nodes	McLag	LACP	NIC0:1 FE, NIC2:1 FE	FE		105-flex-node-mgmt-L3, 140-pfmc-sds-mgmt-L3, 150-flex-stor-mgmt-L3
3	PFMC_nodes	McLag	LACP	NIC0:2 BE, NIC2:2 BE	BE		103-flex-vcsa-ha-L2, 141-pfmc-sds-data1-L2, 142-pfmc-sds-data2-L2, 143-pfmc-vmotion-L2
4	CO_nodes	McLag	LACP	NIC0:1 B0, NIC3:1 B0	B0		105-flex-node-mgmt-L3, 106-flex-vmotion-L2
5	CO_nodes	McLag	LACP	NIC0:2 B1, NIC3:2 B1	B1		151-flex-data1-L2, 152-flex-data2-L2
6	SO_nodes	McLag	LACP	NIC0:1 B0, NIC3:1 B0	B0		150-flex-stor-mgmt-L3, 151-flex-data1-L2, 161-flex-repl1-L3
7	SO_nodes	McLag	LACP	NIC0:2 B1, NIC3:2 B1	B1		152-flex-data2-L2, 162-flex-repl2-L3
8							

Figure 19 ClusterProfilesBonds tab

- A. **Cluster Name** – Specify the cluster name from the [ClusterProfiles](#) tab. Each node has two bonds, so two rows are used for each cluster.
- B. **Type** – This cell must be **McLag** for this deployment.
- C. **Mode** – This cell must be **LACP** for this deployment.
- D. **NICs** – Use a comma-separated list of NIC names that are in the bond. The names must exactly match the names that are entered on the [ClusterProfilesNics](#) tab.
- E. **Bond Name** – Enter a name for the bond. Use a unique name for each bond in a cluster. The same bond names may be used in other clusters.
- F. **Untagged Network** – You can specify one untagged network for each bond. The network name must match the network name that is entered on the [Networks](#) tab. Untagged networks are not used in this deployment example.
- G. **Tagged Networks** – Use a comma-separated list of networks that are in the bond. The network names must match the network names that are entered on the [Networks](#) tab.

5.4.11 Devices tab

The **Devices** tab uses one row for every PowerFlex node in the topology. Add or delete rows as needed to match the number of PowerFlex nodes used in your deployment.

	A	B	C
1	Cluster Name	Device Name	Hostname
2	PFCM_nodes	PFCM1	
3	PFCM_nodes	PFCM2	
4	PFCM_nodes	PFCM3	
5	CO_nodes	CO1	
6	CO_nodes	CO2	
7	CO_nodes	CO3	
8	CO_nodes	CO4	
9	CO_nodes	CO5	
10	CO_nodes	CO6	
11	SO_nodes	SO1	
12	SO_nodes	SO2	
13	SO_nodes	SO3	
14	SO_nodes	SO4	
15	SO_nodes	SO5	
16	SO_nodes	SO6	
17	SO_nodes	SO7	
18	SO_nodes	SO8	

Figure 20 Devices tab

- A. **Cluster Name** – Enter the cluster name from the [ClusterProfiles tab](#) that the PowerFlex node belongs to.
- B. **Device Name** – Enter a unique device name for the PowerFlex node. This cell can be the hostname if column C is not used.
- C. **Hostname (optional)** – If used, the hostname is displayed in the SFM UI device listing along with the **Device Name**.

5.4.12 DeviceNics tab

The **DeviceNics** tab uses one row for every PowerFlex node NIC port that is connected to a leaf switch. Since each PowerFlex node uses four NICs, four rows are used for each node. Add or delete rows as needed to match the number of PowerFlex node NIC ports used in your deployment.

	A	B	C	D	E
1	Cluster Name	Device Name	NIC Name	Node Reference Key	Interface Name
2	PFCM_nodes	PFCM1	NIC0:1 FE	leaf1a	Eth1/9
3	PFCM_nodes	PFCM1	NIC2:1 FE	leaf1b	Eth1/9
4	PFCM_nodes	PFCM1	NIC2:2 BE	leaf1a	Eth1/13
5	PFCM_nodes	PFCM1	NIC0:2 BE	leaf1b	Eth1/13
6	PFCM_nodes	PFCM2	NIC0:1 FE	leaf1a	Eth1/10
7	PFCM_nodes	PFCM2	NIC2:1 FE	leaf1b	Eth1/10
8	PFCM_nodes	PFCM2	NIC2:2 BE	leaf1a	Eth1/14
9	PFCM_nodes	PFCM2	NIC0:2 BE	leaf1b	Eth1/14
10	PFCM_nodes	PFCM3	NIC0:1 FE	leaf1a	Eth1/11
11	PFCM_nodes	PFCM3	NIC2:1 FE	leaf1b	Eth1/11
12	PFCM_nodes	PFCM3	NIC2:2 BE	leaf1a	Eth1/15
13	PFCM_nodes	PFCM3	NIC0:2 BE	leaf1b	Eth1/15
14	CO_nodes	CO1	NIC0:1 B0	leaf1a	Eth1/3
15	CO_nodes	CO1	NIC3:1 B0	leaf1b	Eth1/3
16	CO_nodes	CO1	NIC3:2 B1	leaf1a	Eth1/7
17	CO_nodes	CO1	NIC0:2 B1	leaf1b	Eth1/7
18	CO_nodes	CO2	NIC0:1 B0	leaf1a	Eth1/4
19	CO_nodes	CO2	NIC3:1 B0	leaf1b	Eth1/4
20	CO_nodes	CO2	NIC3:2 B1	leaf1a	Eth1/8
21	CO_nodes	CO2	NIC0:2 B1	leaf1b	Eth1/8
22	CO_nodes	CO3	NIC0:1 B0	leaf2a	Eth1/9
23	CO_nodes	CO3	NIC3:1 B0	leaf2b	Eth1/9
24	CO_nodes	CO3	NIC3:2 B1	leaf2a	Eth1/13
25	CO_nodes	CO3	NIC0:2 B1	leaf2b	Eth1/13
26	CO_nodes	CO4	NIC0:1 B0	leaf2a	Eth1/10

Figure 21 DeviceNics tab

- A. **Cluster Name** – Enter the cluster name that the PowerFlex node belongs to. It must match a cluster name on the [ClusterProfiles tab](#)
- B. **Device Name** – Enter the PowerFlex node device name. It must match the name on the [Devices tab](#).
- C. **NIC Name** – Enter the NIC name from the [ClusterProfilesNics tab](#). Each device in this example has four unique NIC names.
- D. **Node Reference Key** – Enter the reference key that identifies the leaf switch that the PowerFlex NIC is connected to. The reference key must be identical to the key used on the [Nodes tab](#).
- E. **Interface Name** – Enter the interface name that identifies the leaf switch port that the PowerFlex NIC is connected to. This cell must be in the format *Eth1/port_number*.

5.5 Import the blueprint and create the infrastructure

Note: If you have any issues importing the blueprint, see the *Blueprint Errors* section of the *SFM User Guide*.

To import the blueprint into SFM, do the following:

1. In the SFM UI, go to **Blueprints > Blueprint Catalog** and click **IMPORT**.
2. In the **Import template** dialog box, click **Choose Files**.
3. Browse to the blueprint XLSX file and click **Open**. The blueprint filename is displayed in the dialog box.

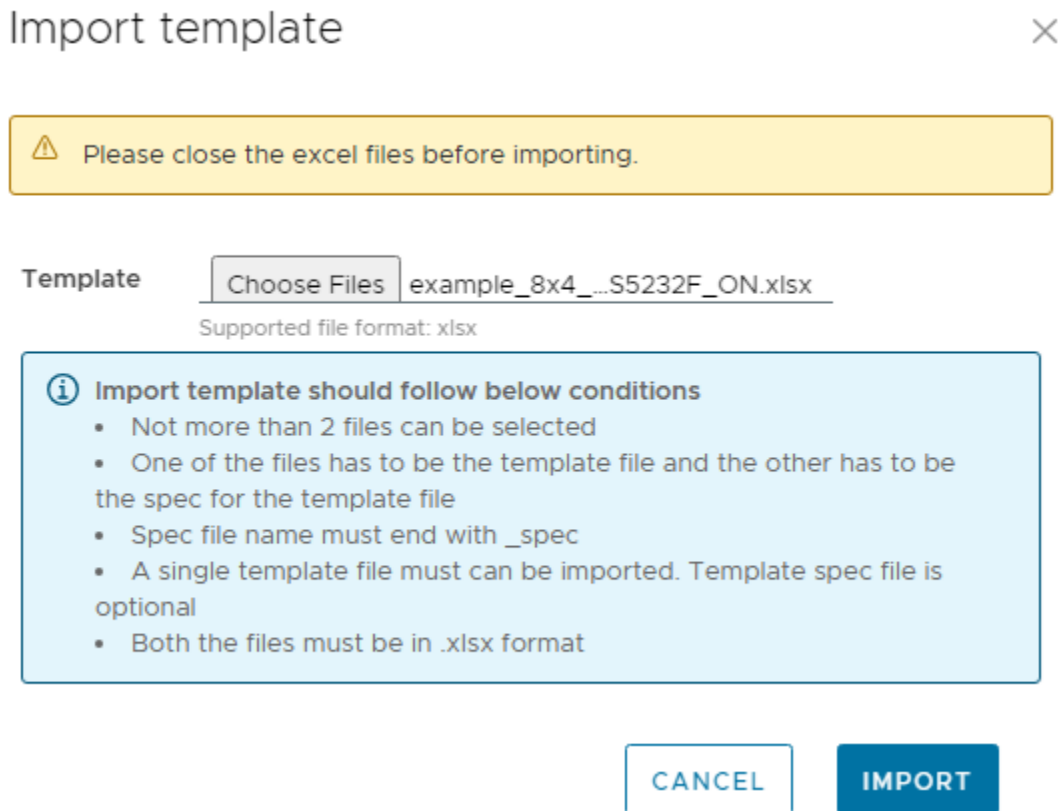


Figure 22 Import template dialog box

- Click **IMPORT**. A success message is displayed at the top of the screen, and the blueprint displays in the list with the **Type** set to **UserDefined**.

Blueprints

[Blueprint Catalog](#)
[Infrastructures](#)

Success: Blueprint: e95ad9dc-fe62-4f18-a57d-c2bab1d2f854 created successfully.

REFRESH
CREATE INFRASTRUCTURE
EXPORT
IMPORT
DELETE

	Name	Blueprint Type	Type	No. of Switches
☐ >>	Pod based 3Tier	Access	PreCanned	16 ⓘ
☐ >>	8x4 Leaf Z9432F-ON spine Z9664F-ON	Access	PreCanned	12 ⓘ
☐ >>	2x2 Leaf Z9432F-ON spine Z9664F-ON	Access	PreCanned	4 ⓘ
☐ >>	Pod based 3Tier	Converged Infrastructure	PreCanned	16 ⓘ
☐ >>	Example 8x4 PowerFlex leaf S5248F-ON and spine S5232F-ON	Converged Infrastructure	UserDefined	12 ⓘ
☐ >>	8x4 PowerFlex leaf S5248F-ON and spine S5232F-ON	Converged Infrastructure	PreCanned	12 ⓘ

Figure 23 Blueprint successfully imported

- On the **Blueprint Catalog** tab, check the box next to the blueprint name and click **CREATE INFRASTRUCTURE**.
- In the **Create Infrastructure** dialog box, provide an infrastructure name, as shown.

Create Infrastructure

Name

PowerFlex Infrastructure

CANCEL
SUBMIT

Figure 24 Create Infrastructure dialog box

7. Click **SUBMIT**. The **Infrastructure Deploy** page displays.

The screenshot displays the 'Infrastructure Deploy' page for a 'PowerFlex Infrastructure'. The page is divided into several sections:

- Top Bar:** Includes a '< BACK' button, the page title 'Infrastructure Deploy (PowerFlex Infrastructure)', a 'Blueprint' dropdown menu showing 'Example 8x4 PowerFlex I...', and a 'VIEW TOPOLOGY' button.
- Actions:** 'EXPORT' and 'IMPORT' buttons are located at the top right.
- Configuration Tabs:** 'Switches', 'Devices', and 'Resources' tabs are present. The 'Switches' tab is active.
- Pod Configuration:** A 'Select Pod' dropdown (with a note 'For Leaf and Spine only') and a 'Select Rack' dropdown (with a note 'For Leaf only') are shown. Below them are two expandable lists: 'Leaf (8)' and 'Spine (4)'.
- Border Leaf Racks Configuration:** A 'Select Rack' dropdown (with a note 'For Border Leaf only') is shown. Below it are two expandable lists: 'Border Leaf (0)' and 'Super Spine (0)'.
- Buttons:** 'VALIDATE' and 'DEPLOY' buttons are at the bottom left.
- Infrastructure Details Panel (Right):**
 - Fabric Settings:** Includes fields for Name (PF-Fabric), Fabric Type (VxLan), Topology Type (RackBasedClos), STP Edge (Enable), LACP Fallback Mode (Fallback), VTEP CIDR (172.30.0.0/16), BGP CIDR (172.16.128.0/24), Leaf ASN (65011), Spine ASN (65012), Super Spine ASN (65013), and Border Leaf ASN (65014).
 - Management Settings:** Includes fields for NTP (100.67.10.20), DNS (100.67.10.1), Sflow (Not Configured), and Syslog (Not Configured).

Figure 25 Infrastructure Deploy page

Some information that is imported from the blueprint XLSX file is shown under **Infrastructure Details** on the far right of the page. There are also tabs named **Switches**, **Devices**, and **Resources** that contain the remainder of the imported information.

Note: If you leave the **Infrastructure Deploy** page, you can return to it by doing the following:

1. Click **Blueprints > Infrastructures**.
2. Click the > icon next to the infrastructure name to expand it.
3. Click the radio button next to the blueprint name
4. Click **EDIT & DEPLOY** to return to the **Infrastructure Deploy** page.

- Click the > icon next to **Leaf** to expand the list of leaf switches. The page displays the list of leaf switches that are imported from the blueprint.

EXPORT
IMPORT

Switches
Devices
Resources

Pods Select Pod ▾
For Leaf and Spine only

Racks Select Rack ▾
For Leaf only

▼
Leaf (8)

ADD RACK
X DELETE RACK
CONFIG MC-LAG
EDIT SWITCH

		IP Address ▾	Node Role ▾	Pod Name ▾	Name ↑ ▾	Status ▾
○	»	100.67.124.189	Leaf	1	leaf1a	NOT_DEPLOYED
○	»	100.67.124.190	Leaf	1	leaf1b	NOT_DEPLOYED
○	»	100.67.124.187	Leaf	1	leaf2a	NOT_DEPLOYED
○	»	100.67.124.188	Leaf	1	leaf2b	NOT_DEPLOYED
○	»	100.67.124.191	Leaf	1	leaf3a	NOT_DEPLOYED
○	»	100.67.124.192	Leaf	1	leaf3b	NOT_DEPLOYED
○	»	100.67.124.193	Leaf	1	leaf4a	NOT_DEPLOYED
○	»	100.67.124.194	Leaf	1	leaf4b	NOT_DEPLOYED

Nodes per page
10 ▾
1 - 8 of 8 Nodes

Figure 26 Switches tab – leaf switches

- Click the > icon next to **Spine** to expand the list of spine switches. The page displays the list of spine switches that are imported from the blueprint.

Switches

Devices

Resources

EXPORT

IMPORT

ADD SPINE

DELETE SPINE

ADD RACK

DELETE RACK

Racks

Select Rack

Spine (4)

		IP Address		Node Role		Name		Status	
	>>	100.67.124.196		Spine		spine4		NOT_DEPLOYED	
	>>	100.67.124.185		Spine		spine1		NOT_DEPLOYED	
	>>	100.67.124.195		Spine		spine3		NOT_DEPLOYED	
	>>	100.67.124.186		Spine		spine2		NOT_DEPLOYED	

Nodes per page 10 1 - 4 of 4 Nodes

> Leaf (8)

Figure 27 Switches tab - spine switches

10. Near the top of the page, click the **Devices** tab to view the list of PowerFlex nodes imported from the blueprint.

Switches

Devices

Resources

EXPORT

IMPORT

▼ Devices

EDIT

DELETE

☐	Name	Namespace	Cluster Name	Status
☐	CO1	Converged Infrastructure	CO_nodes	NOT_DEPLOYED
☐	CO2	Converged Infrastructure	CO_nodes	NOT_DEPLOYED
☐	CO3	Converged Infrastructure	CO_nodes	NOT_DEPLOYED
☐	CO4	Converged Infrastructure	CO_nodes	NOT_DEPLOYED
☐	CO5	Converged Infrastructure	CO_nodes	NOT_DEPLOYED
☐	CO6	Converged Infrastructure	CO_nodes	NOT_DEPLOYED
☐	PPMC1	Converged Infrastructure	PPMC_nodes	NOT_DEPLOYED
☐	PPMC2	Converged Infrastructure	PPMC_nodes	NOT_DEPLOYED
☐	PPMC3	Converged Infrastructure	PPMC_nodes	NOT_DEPLOYED
☐	SO1	Converged Infrastructure	SO_nodes	NOT_DEPLOYED

Manage Columns

Devices per page 10

1 - 10 of 29 Devices

1 / 3

> Clusters

> Downlinks

Figure 28 Devices tab - PowerFlex nodes

11. Below the device list, click the > icon next to **Clusters** to view the list of imported clusters.

Switches

Devices

Resources

EXPORT

IMPORT

> Devices

< Clusters

EDIT

DELETE

☐		Name	Tenant Id	Status
☐	>>	SO_nodes	PF_VRF	NOT_DEPLOYED
☐	>>	PFMC_nodes	PF_VRF	NOT_DEPLOYED
☐	>>	CO_nodes	PF_VRF	NOT_DEPLOYED

Clusters per page 10 1 - 3 of 3 Clusters

> Downlinks

Figure 29 Devices tab – clusters

12. Click the **Resources** tab to view the imported networks.

[< BACK](#)

Infrastructure Deploy
(PowerFlex Infrastructure)

Blueprint [Power-Flex Converged Inf...](#)

[VIEW TOPOLOGY](#)

[Switches](#)

[Devices](#)

[Resources](#)

[EXPORT](#)

[IMPORT](#)

▼ Networks

[EDIT](#)

[X DELETE](#)

☐	Network Name	Network Type	VLAN Id	Status
☐	103-flex-vcsa-ha-L2	VxLan	103	NOT_DEPLOYED
☐	105-flex-node-mgmt-L3	VxLan	105	NOT_DEPLOYED
☐	106-flex-vmotion-L2	VxLan	106	NOT_DEPLOYED
☐	140-pfmc-sds-mgmt-L3	VxLan	140	NOT_DEPLOYED
☐	141-pfmc-sds-data1-L2	VxLan	141	NOT_DEPLOYED
☐	142-pfmc-sds-data2-L2	VxLan	142	NOT_DEPLOYED
☐	143-pfmc-vmotion-L2	VxLan	143	NOT_DEPLOYED
☐	150-flex-stor-mgmt-L3	VxLan	150	NOT_DEPLOYED
☐	151-flex-data1-L2	VxLan	151	NOT_DEPLOYED
☐	152-flex-data2-L2	VxLan	152	NOT_DEPLOYED

Networks per page 10 1 - 10 of 12 Networks 1 / 2

> Tenants

Figure 30 Resources tab – networks

13. At the bottom of the page, click the > icon next to **Tenants** to view the imported PowerFlex VRF.

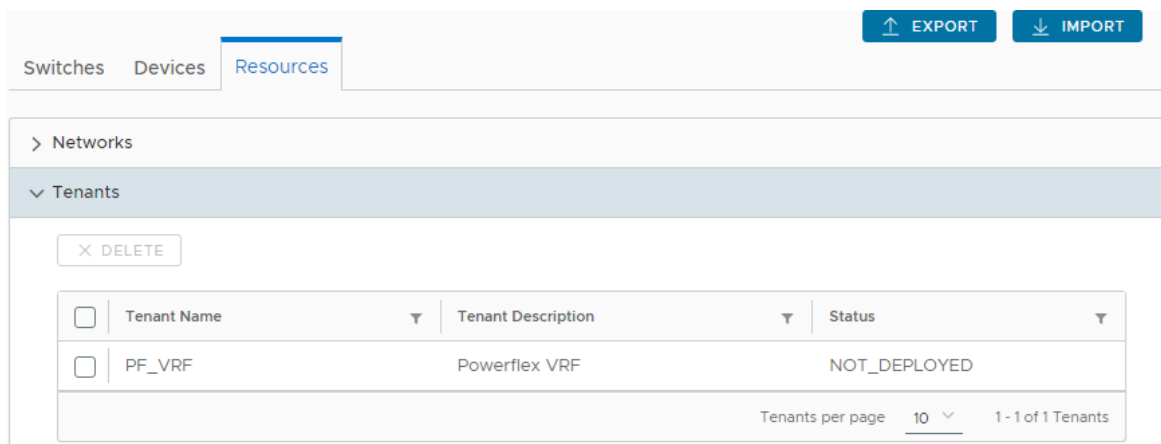


Figure 31 Resources tab – PowerFlex VRF

5.6 Validate the infrastructure

During validation, SFM verifies that it can connect to each switch and checks the SONiC operating system version and switch model to ensure that they are compatible with SFM. SFM also checks the validity of the service tag and MAC address on each switch if these items were specified in the blueprint file.

Note: See the *SFM User Guide* for more information about the checks made during the validation process.

To validate the infrastructure, do the following:

1. At the bottom of the **Infrastructure Deploy** page, click **VALIDATE**.

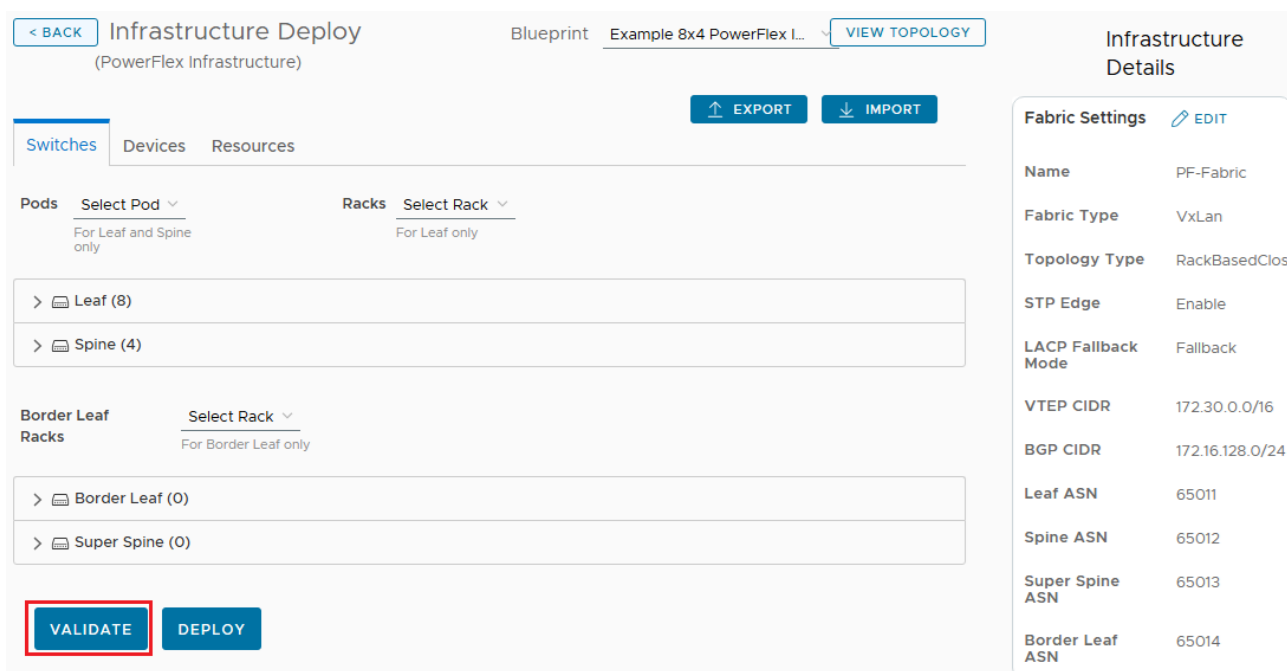


Figure 32 Validate button

A message indicating **Validation in Progress** is displayed in the **Action Status** column.

Blueprints

Blueprint Catalog **Infrastructures**

REFRESH

PowerFlex Infrastructure

DELETE

EDIT & DEPLOY

VIEW

Blueprint Name	Blueprint Type	Recent Action	Action Status
Example 8x4 PowerFlex leaf S5248F-ON and spine S5232F-ON	Converged Infrastructure	Validate	Validation Inprogress ⓘ

Blueprints per page 10 1 - 1 of 1 Blueprints

Figure 33 Validation in progress

2. Periodically click **REFRESH** to update the status. Validation may take a minute or two.

When validation is complete, the **Action Status** column displays a message indicating either **Validation Succeeded** or **Validation Failed**.

Blueprints

Blueprint Catalog **Infrastructures**

REFRESH

PowerFlex Infrastructure

DELETE

EDIT & DEPLOY

VIEW

Blueprint Name	Blueprint Type	Recent Action	Action Status
Example 8x4 PowerFlex leaf S5248F-ON and spine S5232F-ON	Converged Infrastructure	Validate	Validation Succeeded ⓘ

Blueprints per page 10 1 - 1 of 1 Blueprints

Figure 34 Validation succeeded

EDIT & DEPLOY

VIEW

Blueprint Name	Blueprint Type	Recent Action	Action Status
	Converged Infrastructure	Validate	Validation Failed ⚠

Blueprints per page 10 1 - 1 of 1 Blueprints

Figure 35 Validation failed

3. If validation succeeded, go to the [Deploy the infrastructure](#) section. If validation failed, click the  icon in the **Action Status** column to open **Action Messages** box. Example error messages are shown below.

Action Messages

S.No	Message(s)
1	Add Switch Error: Backend Service Failed: Error Code: SVIF0101. Switch IP address already registered. Switch leaf3a 100.67.128.30 already registered.
2	Add Switch Error: Backend Service Failed: Error Code: SVIF0101. Switch IP address already registered. Switch leaf3b 100.67.128.130 already registered.

CLOSE

Figure 36 Action Messages

4. Address any issues that are listed in the **Action Messages** box. Information on the **Infrastructure Deploy** page and tabs can be modified to address the issues using the edit buttons provided.
5. After addressing the issues, run **VALIDATE** again. Repeat as needed until validation succeeds.

5.7 Deploy the infrastructure

Note: If you leave the **Infrastructure Deploy** page, you can return to it by doing the following:

1. Click **Blueprints > Infrastructures**.
2. Click the > icon next to the PowerFlex infrastructure name to expand it.
3. Click the radio button next to the blueprint name.
4. Click **EDIT & DEPLOY** to return to the **Infrastructure Deploy** page.

Deploying the infrastructure configures all leaf and spine switches in the fabric. To deploy the infrastructure, do the following:

1. At the bottom of the **Infrastructure Deploy** page, click **DEPLOY**. A confirmation dialog box displays.

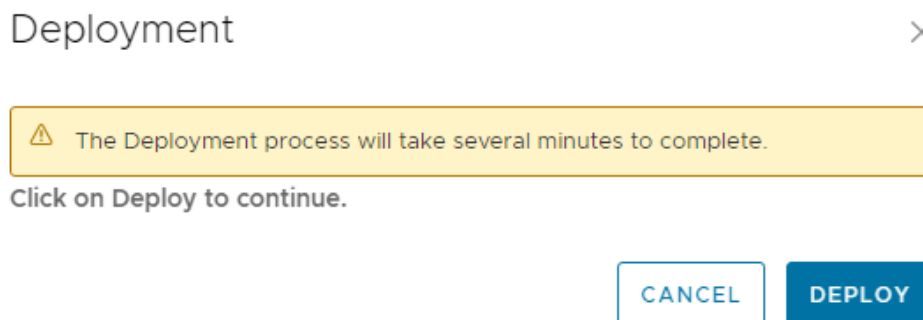


Figure 37 Deployment confirmation

- Click **DEPLOY** to continue. A message indicating **Deployment in progress** displays in the **Action Status** column.

Blueprints

Blueprint Catalog **Infrastructures**

REFRESH **EDIT & DEPLOY** **DELETE**

Infrastructure Name	Blueprint Name	Blueprint Type	Recent Action	Action Status
PowerFlex Infrastructure	Power-Flex Converged Infrastructure Blueprint	Converged Infrastructure	Deploy	Deployment Inprogress

Blueprints per page 10 1 - 1 of 1 Blueprints

Figure 38 Deployment in progress

Note: While deployment is in progress, you can hover your mouse pointer over the icon to view status updates.

- Periodically click the **REFRESH** button to update the status. Deployment may take about 5 minutes.

When the process is complete, a message indicating either **Deployment Succeeded** or **Deployment Failed** is displayed in the **Action Status** column.

Blueprints

Blueprint Catalog **Infrastructures**

REFRESH

PowerFlex Infrastructure **DELETE**

EDIT & DEPLOY **VIEW**

Blueprint Name	Blueprint Type	Recent Action	Action Status
Example 8x4 PowerFlex leaf S5248F-ON and spine S5232F-ON	Converged Infrastructure	Deploy	Deployment Succeeded

Blueprints per page 10 1 - 1 of 1 Blueprints

Figure 39 Deployment succeeded

EDIT & DEPLOY **VIEW**

Blueprint Name	Blueprint Type	Recent Action	Action Status
	Converged Infrastructure	Deploy	Deployment Failed

Blueprints per page 10 1 - 1 of 1 Blueprints

Figure 40 Deployment failed

4. If deployment succeeded, go to the [Verify the](#) section. If deployment failed, click the  icon in the **Action Status** column to open **Action Messages** box.
5. Address any issues that may be listed in the **Action Messages** box. Information on the **Infrastructure Deploy** tabs can be changed if needed to address any issues using the edit buttons provided.
6. After addressing the issues, run **DEPLOY** again. Repeat as needed until deployment succeeds.

5.8 Verify the fabric

After the infrastructure is deployed, it is recommended to verify that the fabric is configured to your expectations. To verify the fabric, complete the following steps:

1. In the SFM UI, click **Fabric** in the left pane. The **Managed Fabrics** page opens in the right pane.

Managed Fabrics

REFRESH
EDIT
DELETE

	Fabric Name	No. of Leaves	No. of Spines	No. of SuperSpines	No. of BorderLeaves	Fabric Type	Site	Action
☐	PF-Fabric	8	4	0	0	VxLan	1	 MANAGE

Fabrics per page 10 1 - 1 of 1 Fabrics

Figure 41 Managed Fabrics page

2. Next to the PowerFlex fabric name, click **MANAGE**. The **Switches** tab displays.

< Fabrics / Fabric: PF-Fabric

Switches	LAG	Topology	Fabric Upgrade	Downlinks	Uplinks	Health	Settings
----------	-----	----------	----------------	-----------	---------	--------	----------

REFRESH	DELETE	CLAIM OWNERSHIP	RUN CLI COMMANDS
---------	--------	-----------------	------------------

☐	Name	Role	IP Address	Status	Mac Address
☐ >>	leaf1a	Leaf	100.67.124.189	Online	B0:4F:13:2B:6A:A0
☐ >>	leaf1b	Leaf	100.67.124.190	Online	8C:47:BE:AD:23:0F
☐ >>	leaf2a	Leaf	100.67.124.187	Online	B0:4F:13:2B:89:A0
☐ >>	leaf2b	Leaf	100.67.124.188	Online	B0:4F:13:2B:5E:A0
☐ >>	leaf3a	Leaf	100.67.124.191	Online	B0:4F:13:37:09:C0
☐ >>	leaf3b	Leaf	100.67.124.192	Online	B0:4F:13:37:0F:C0
☐ >>	leaf4a	Leaf	100.67.124.193	Online	B0:4F:13:36:50:C0
☐ >>	leaf4b	Leaf	100.67.124.194	Online	B0:4F:13:36:DF:40
☐ >>	spine1	Spine	100.67.124.185	Online	B0:4F:13:9D:CD:20
☐ >>	spine2	Spine	100.67.124.186	Online	B0:4F:13:9D:6E:20
☐ >>	spine3	Spine	100.67.124.195	Online	8C:04:BA:A9:46:40
☐ >>	spine4	Spine	100.67.124.196	Online	3C:2C:30:72:74:00

Nodes per page 20 1 - 12 of 12 Nodes

Figure 42 Switches tab

3. Ensure that all leaf and spine switches in the fabric are listed and are **Online**.
4. Click the >> icon next to a switch name to view more information. The right pane contains tabs that are named **Switch Details**, **Interfaces**, and **Health**, outlined in red in the figure below.

< Fabrics / Fabric: PF-Fabric

Switches LAG Topology Fabric Upgrade Downlinks Uplinks Health Settings

REFRESH DELETE CLAIM OWNERSHIP RUN CLI COMMANDS

Name	Switch Details	Interfaces	Health
leaf1a	<<	>>	>>
leaf1b	<<	>>	>>
leaf2a	<<	>>	>>
leaf2b	<<	>>	>>
leaf3a	<<	>>	>>
leaf3b	<<	>>	>>

REFRESH EDIT

Name	Admin Status	Operational Status	MTU	Speed (Mbps)
Eth1/1	up	down	9216	25000
Eth1/2	up	down	9216	25000
Eth1/3	up	down	9216	25000
Eth1/4	up	down	9216	25000

Figure 43 Switch Details, Interfaces, and Health tabs

On the **Interfaces** tab, the admin status, operational status, and other settings for each port are shown.

5. You can select and edit individual ports to change settings such as admin state, MTU, FEC, and other settings.

Note: There is no need to change settings on ports that are connected to PowerFlex nodes or other switches in the fabric, unless you are doing troubleshooting.

6. Click the **LAG** tab at the top of the **Fabrics** page. All LAGs in the topology are listed. The list may cover several pages for larger deployments.

Switches	LAG	Topology	Fabric Upgrade	Downlinks	Uplinks	Health	Settings
REFRESH + ADD EDIT DELETE							
		Name	Info	Lag Type/ Mode	Admin State		
☐	>>	CO_nodes_CO1_B0	Interfaces 2	McLag LACP	✓ up		
☐	>>	CO_nodes_CO1_B1	Interfaces 2	McLag LACP	✓ up		
☐	>>	CO_nodes_CO2_B0	Interfaces 2	McLag LACP	✓ up		
☐	>>	CO_nodes_CO2_B1	Interfaces 2	McLag LACP	✓ up		
☐	>>	CO_nodes_CO3_B0	Interfaces 2	McLag LACP	✓ up		
☐	>>	CO_nodes_CO3_B1	Interfaces 2	McLag LACP	✓ up		
☐	>>	CO_nodes_CO4_B0	Interfaces 2	McLag LACP	✓ up		
☐	>>	CO_nodes_CO4_B1	Interfaces 2	McLag LACP	✓ up		
☐	>>	CO_nodes_CO5_B0	Interfaces 2	McLag LACP	✓ up		
☐	>>	CO_nodes_CO5_B1	Interfaces 2	McLag LACP	✓ up		
LAGs per page 10 1 - 10 of 58 LAGs < 1 / 6 >							

Figure 44 LAG tab

- a. Verify that all PowerFlex node LAGs have two interfaces, are configured for **MC-LAG** and **LACP**, and the **Admin State** is **up** as shown above.
- b. Click the >> icon next to a LAG name and then click the > icon next to **Interfaces**. More details are shown such as the switch and ports that are used in the LAG.

LAG Details					✕	
LAG ID		SO_nodes_SO18_B0				
▼ Interfaces 2						
		Switch Name	Interface	PortChannel Id	Operational State	
		leaf3a	Eth1/6	250	down	
		leaf3b	Eth1/6	250	down	
Lag Type / Mode		McLag LACP				
Rack ID		4cc0ccc5-cf12-4a05-bce2-e1b87c84c14c				
Fabric ID		df501624-e8db-4a96-bf83-e558107e9caa				

Figure 45 LAG details

7. Click the **Topology** tab at the top of the **Fabrics** page. The fabric topology and switch hostnames are shown.

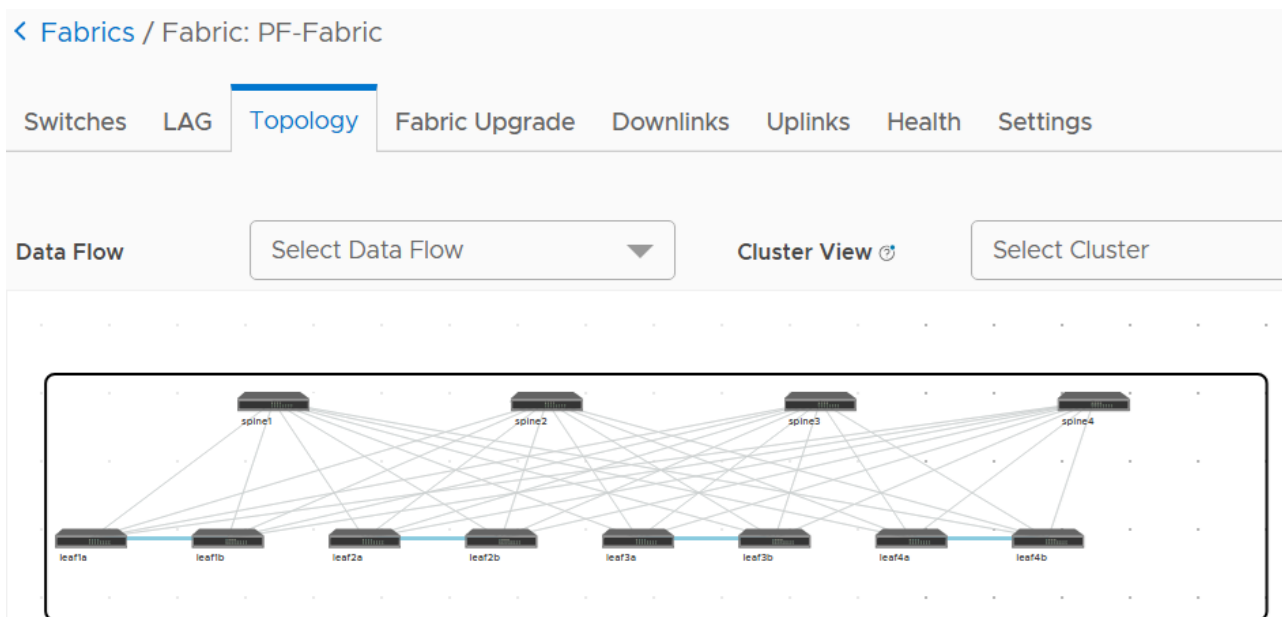


Figure 46 Topology tab

- a. Ensure that all leaf-leaf, and leaf-spine connections are shown. A missing connection indicates that a link is down or not physically connected.
- b. You can view switch details such as IP address and service tag by hovering your mouse pointer over a switch. You can view link details by hovering your mouse pointer over a link.

8. Click the **Downlinks** tab at the top of the **Fabrics** page. Downlinks are the connections to the PowerFlex nodes. Each LAG is a downlink, so there should be two downlinks for every PowerFlex node.

Switches

LAG

Topology

Fabric Upgrade

Downlinks

Uplinks

Health

Settings

REFRESH

+ ADD

EDIT

DELETE

	Name	Info	Tenant Name	Status
>>	CO_nodes_CO_nodes_CO1_B0_pf_vrf	Networks 2 Interfaces 1	PF_VRF	Success
>>	CO_nodes_CO_nodes_CO1_B1_pf_vrf	Networks 2 Interfaces 1	PF_VRF	Success
>>	CO_nodes_CO_nodes_CO2_B0_pf_vrf	Networks 2 Interfaces 1	PF_VRF	Success
>>	CO_nodes_CO_nodes_CO2_B1_pf_vrf	Networks 2 Interfaces 1	PF_VRF	Success
>>	CO_nodes_CO_nodes_CO3_B0_pf_vrf	Networks 2 Interfaces 1	PF_VRF	Success
>>	CO_nodes_CO_nodes_CO3_B1_pf_vrf	Networks 2 Interfaces 1	PF_VRF	Success
>>	CO_nodes_CO_nodes_CO4_B0_pf_vrf	Networks 2 Interfaces 1	PF_VRF	Success
>>	CO_nodes_CO_nodes_CO4_B1_pf_vrf	Networks 2 Interfaces 1	PF_VRF	Success
>>	CO_nodes_CO_nodes_CO5_B0_pf_vrf	Networks 2 Interfaces 1	PF_VRF	Success
>>	CO_nodes_CO_nodes_CO5_B1_pf_vrf	Networks 2 Interfaces 1	PF_VRF	Success

Downlinks per page

10

1 - 10 of 58 Downlinks

1

6

Figure 47 Downlinks tab

- a. Click the >> icon next to a downlink name to view additional details. The **Downlink Details** pane displays on the right.
- b. Click the > icon next to **Tagged Networks**. The list shows the tagged networks that the downlink is in.

Switches

LAG

Topology

Fabric Upgrade

Downlinks

Uplinks

Health

Settings

REFRESH

+ ADD

EDIT

DELETE

	Name
◀	CO_nodes_CO_nodes_CO1_B0_pf_vrf
▶	CO_nodes_CO_nodes_CO1_B1_pf_vrf
▶	CO_nodes_CO_nodes_CO2_B0_pf_vrf
▶	CO_nodes_CO_nodes_CO2_B1_pf_vrf
▶	CO_nodes_CO_nodes_CO3_B0_pf_vrf
▶	CO_nodes_CO_nodes_CO3_B1_pf_vrf
▶	CO_nodes_CO_nodes_CO4_B0_pf_vrf
▶	CO_nodes_CO_nodes_CO4_B1_pf_vrf
▶	CO_nodes_CO_nodes_CO5_B0_pf_vrf
▶	CO_nodes_CO_nodes_CO5_B1_pf_vrf

Downlink Details

Downlink Id

CO_nodes_CO_nodes_CO1_B0_pf_vrf

Tenant Id

pf_vrf

Tenant Name

PF_VRF

Tagged Networks

Name	VLAN
105-flex-node-mgmt-L3	105
106-flex-vmotion-L2	106

Untagged Network

Interfaces

CO_nodes_CO1_B0

Status

Success

Figure 48 Downlink details

9. Click the **Health** tab to display the fabric health.

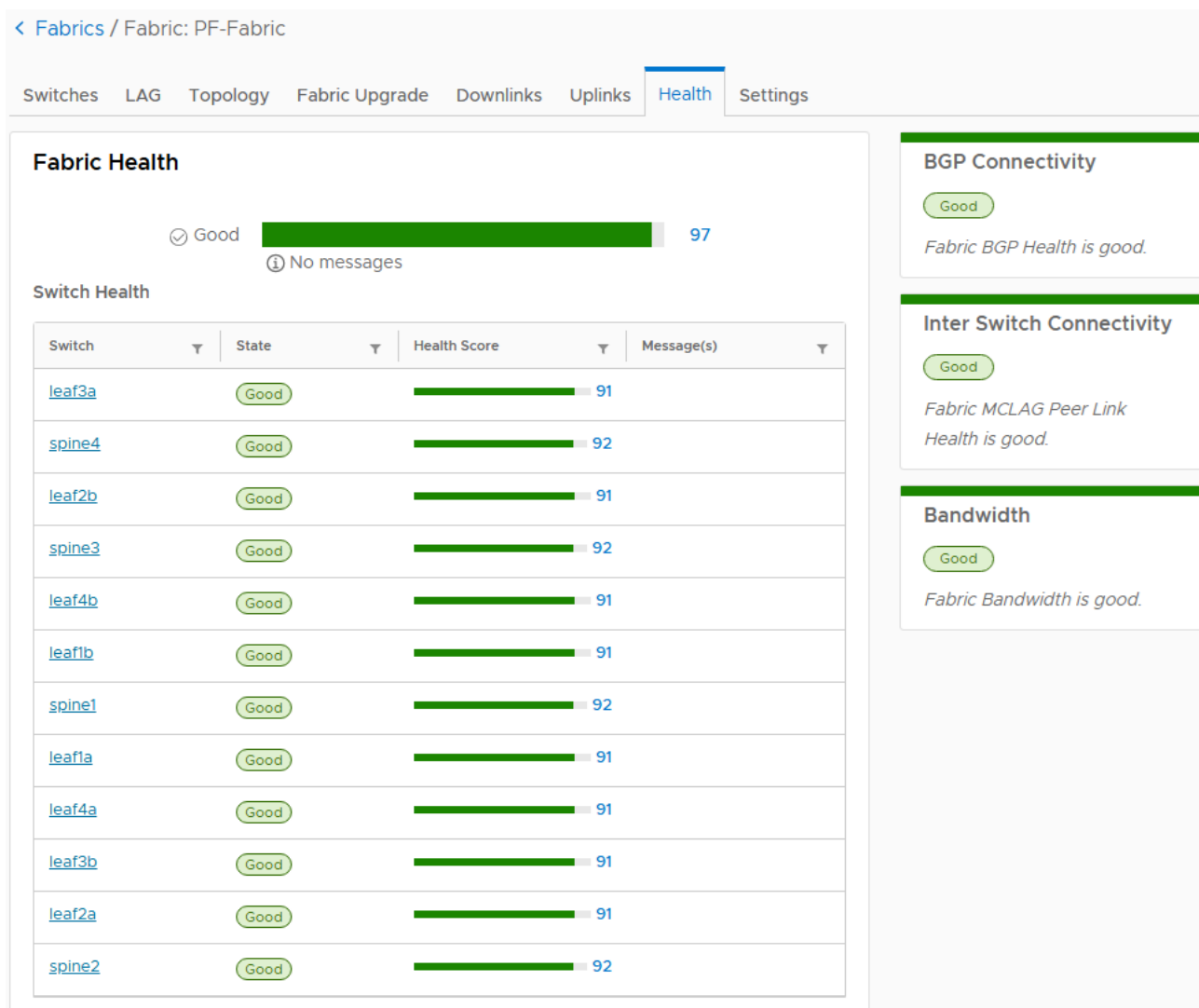


Figure 49 Health tab

10. Ensure that the **State** is **Good**, and the **Health Score** is green for each switch. If any switches have a yellow or red **Health Score**, see the **Messages** column for the cause.

6 Deploy PowerFlex

After the fabric has been configured and verified, PowerFlex can be deployed according to your PowerFlex deployment and administration documentation.

In PowerFlex Manager, be sure to use a partial network automation template, such as **Compute-Linux-Partial Network** or **Storage-Partial Network** as a starting point.

Note: Using a PowerFlex template that does not include “Partial Network” in the name may result in deployment failures.

A Validated components

The following tables include the hardware, software, and firmware that is used to validate the example in this guide. Other supported hardware, software, and firmware can be used.

A.1 Dell PowerSwitch systems

Table 5 Dell PowerSwitch models and operating systems

Qty	Item	Role	SONiC version
8	Dell PowerSwitch S5248F-ON	Leaf switch	4.4.1 Enterprise Standard
4	Dell PowerSwitch S5232F-ON	Spine switch	4.4.1 Enterprise Standard
4	Dell PowerSwitch S3248T-ON	OOB management switch	4.4.1 Lite

A.2 SONiC SmartFabric Manager

One PowerEdge R640 server running VMware ESXi was used as the host for the SONiC SmartFabric Manager virtual machine. The software that is used is shown below.

Table 6 SmartFabric Manager host software

Item	Version
VMware ESXi	8.0.3 build 24280767 Dell customized A02
SONiC SmartFabric Manager	1.2.0

A.3 PowerFlex nodes

Table 7 PowerFlex R650 S custom node configuration

Qty per node	Item	Firmware version
2	Intel Xeon Gold 6326 CPU @ 2.90 GHz, 16 cores	
8	32 GB DDR4 DIMM (256 GB total)	
5	3576.98 GB SATA SSD (HBA355i Fnt controller)	
2	447.13 GB SATA SSD (BOSS-S2 controller)	
1	Dell HBA355i Fnt Storage Controller	24.15.14.00
1	Boot Optimized Storage Solution (BOSS-S2) Controller	2.5.13.4009
2	Mellanox ConnectX-5 EN 25 GbE dual-port SFP28 adapter	16.35.10.12
-	BIOS	1.14.1
-	CPLD	1.1.1

-	iDRAC with Lifecycle Controller	7.10.50.201
-	Backplane	7.10

Table 8 PowerFlex R650 C custom node configuration

Qty per node	Item	Firmware version
2	Intel Xeon Gold 6326 CPU @ 2.90 GHz, 16 cores	
8	32 GB DDR4 DIMM (256 GB total)	
2	447.13 GB SATA SSD	
1	Boot Optimized Storage Solution (BOSS-S2) Controller	2.5.13.4009
2	Mellanox ConnectX-5 EN 25 GbE dual-port SFP28 adapter	16.35.10.12
-	BIOS	1.14.1
-	CPLD	1.1.1
-	iDRAC with Lifecycle Controller	7.10.50.201

A.4 PowerFlex Management Controller

Three PowerFlex R650 S custom nodes and the software in the table below were used to build the PowerFlex Management Controller 2.0 cluster.

Table 9 PowerFlex Management Controller software

Item	Version
PowerFlex Manager	4.6.0.0 build 1258
VMware ESXi	7.0.3 build 23307199
VMware vCenter Server Appliance	7.0.3 build 22837322

B BGP EVPN with VXLAN overview

BGP EVPN with VXLAN stretches virtual (overlay) Layer 2 networks across a physical (underlay) Layer 3 leaf-spine fabric. This architecture allows for the scalability of Layer 3 networks with the mobility benefits of a Layer 2 network. For example, a VM or physical host can be moved from one rack to another without the need to change its IP address and gateway information. All paths in the VXLAN fabric are active, and the spanning tree protocol does not block any paths. SFM automates the configuration of the VXLAN fabric.

EVPN uses BGP to exchange endpoint MAC and IP address information between VXLAN tunnel endpoints (VTEPs). In this topology, each pair of leaf switches in a rack is one logical VTEP. Packets destined to a remote device are sent to the local VTEP as the first hop. The packet is encapsulated with a VXLAN header that includes the Virtual Network Identifier (VNI) and the IP address of the destination host. See the figure below.

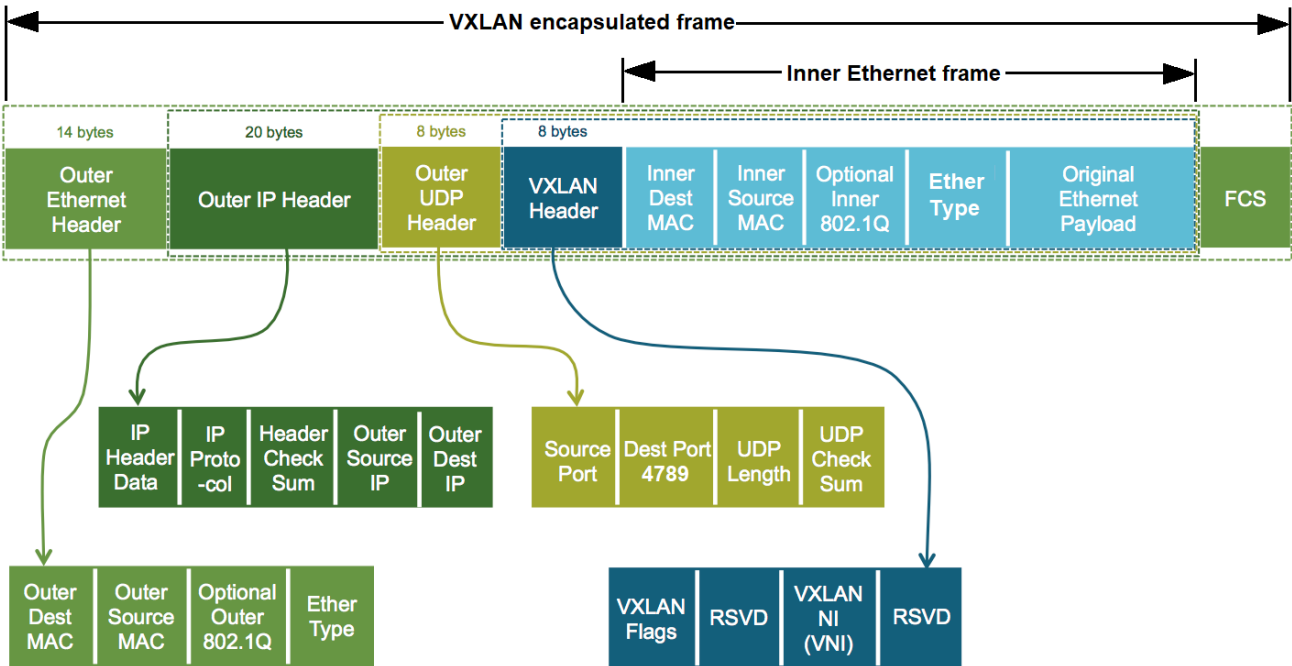


Figure 50 VXLAN encapsulated frame

The destination VTEP extracts the original packet and routes it to the destination. VXLAN encapsulation adds approximately 50 bytes of overhead to each Ethernet frame. As a result, all switches in the underlay (physical) network must be configured to support an MTU of at least 1600 bytes on all participating interfaces.

Note: In this deployment example, switch interfaces are set to their maximum supported MTU size of 9216 bytes for best performance.

Anycast gateways with the same IP address are configured on each leaf pair. The anycast gateway IP address is set as the default gateway for all hosts and VMs on that virtual network. Hosts and VMs with anycast gateways use the same gateway information while behind different leaf pairs. This configuration replaces the Virtual Router Redundancy Protocol (VRRP) and enables hosts and VMs to migrate from one leaf pair to another without changing the network configuration.

C Resources

C.1 Dell Networking resources

[Dell Technologies Networking Solutions Info Hub](#) – This site provides guides to assist with planning and deploying various solutions with Dell Networking.

[Dell Technologies Enterprise SONiC Distribution](#) (account required) – This site includes SONiC user guides and software.

[SmartFabric Manager documentation](#) (account required)

[SmartFabric Manager software](#) (account required)

[Dell PowerSwitch S3248T-ON Documentation](#)

[Dell PowerSwitch S5212F-ON Documentation](#)

[Dell PowerSwitch S5248F-ON Documentation](#)

[Dell PowerSwitch S5232F-ON Documentation](#)

[Dell Networking Transceivers and Cables](#)

C.2 PowerFlex guides

See your sales account representative for these guides.

Dell PowerFlex Appliance with PowerFlex 4.x Network Planning Guide

Dell PowerFlex Appliance with PowerFlex 4.x Deployment Guide

C.3 Technical support

For technical support, go to <http://www.dell.com/support>.